

A Symmetry-Allowed Minimal Model for Two-Dimensional Coherent Terahertz Spectroscopy of TmFeO_3 in the Γ_2 Phase

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We construct a symmetry-allowed minimal Hamiltonian for the rare-earth orthoferrite TmFeO_3 in its low-temperature Γ_2 phase and use it to derive the third-order nonlinear response that produces the observed two-dimensional coherent terahertz spectroscopy (2DCS) peak structure, for the three pump–detect geometries ($H \parallel a, E \parallel c, \text{VV}$), ($H \parallel a, E \parallel c, \text{VH}$), and ($H \parallel c, E \parallel a, \text{VH}$). The model couples a four-sublattice Fe Landau–Lifshitz–Gilbert sector (with Heisenberg, Dzyaloshinskii–Moriya, and biaxial single-ion anisotropy) to a Tm^{3+} non-Kramers crystal-field qutrit $\{|E_1(A_1)\rangle, |E_2(A_1)\rangle, |E_3(A_2)\rangle\}$ projected onto the $\text{SU}(3)$ Gell–Mann basis. We derive Bertaut linear and quadratic spin-reorientation (SR) modes carefully, including the Dzyaloshinskii–Moriya gap on the qAFM mode (which a naive exchange-only treatment misses by orders of magnitude), and we cast the Tm dissipation in CPTP-compliant Lindblad form. Each 2DCS peak is identified with a specific Liouville pathway in $\chi^{(3)}$, with explicit double-sided Feynman labelling, rephasing/non-rephasing assignment, and a formula for the relative amplitude in terms of the projected matrix elements. The derivation closes with an explicit 4×4 linearized magnon matrix M^{Γ_2} and a sympy/numpy verification script that reproduces every algebraic claim.

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I. INTRODUCTION

Rare-earth orthoferrites $R\text{FeO}_3$ are a paradigmatic platform for coherent ultrafast control of antiferromagnetic order. Resonant terahertz excitation of the canted G-type Fe^{3+} sublattice [1, 2] launches the high-frequency quasi-antiferromagnetic (qAFM) and the low-frequency quasi-ferromagnetic (qFM) magnon modes, while resonant excitation of the rare-earth crystal-field (CF) levels modifies the effective Fe anisotropy and drives a coherent spin reorientation [3–5]. TmFeO_3 is a benchmark case because Tm^{3+} is a non-Kramers ion ($J = 6$, 3H_6), its low-energy spectrum consists of three well-separated singlets within the bottom 8 meV ($\epsilon_2 - \epsilon_1 = 1.92$ meV, $\epsilon_3 - \epsilon_1 = 7.84$ meV) that are addressable by single-cycle THz fields, and the compound exhibits a continuous Fe spin reorientation in the ac plane bounded by *two* second-order phase transitions, $T_{\text{SR1}} \approx 81$ K and $T_{\text{SR2}} \approx 91$ K, between the high-temperature Γ_4 and the low-temperature Γ_2 phases, with the mixed Γ_{24} rotation between them [4, 8–11].

Two-dimensional coherent terahertz spectroscopy (2DCS, 2DTS) [22–26] resolves the third-order nonlinear susceptibility $\chi^{(3)}(t_1, t_2, t_3)$ of magnetic materials into individual Liouville pathways. In TmFeO_3 this is the natural language for asking which combinations of the four Fe Bertaut modes ($\mathbf{F}, \mathbf{G}, \mathbf{C}, \mathbf{A}$) and the three Tm CF coherences ($\omega_{12}, \omega_{13}, \omega_{23}$) couple, and through which microscopic vertices.

A. What this paper provides

1. A symmetry-derived minimal Hamiltonian, written in a single, internally-consistent normalization that we audit end to end. The Hamiltonian derivation is collected first, through the uniform $\mathbf{q} = 0$ working model of Sec. VII.
2. A non-Kramers Tm projected-dipole sector with explicit T -symmetric gauge for the three singlets (Sec. III), so that all projected operators $\mathbf{J}_a^{\text{eff}}$ are purely imaginary and the $m_J \rightarrow -m_J$ relative parities of $|E_{1,2}\rangle$ and $|E_3\rangle$ are stated.
3. A dedicated 2DCS framework section (Sec. VIII) that introduces the nonlinear field, response-function language, and the three experimental geometries before any dynamical calculation.
4. A later dynamical analysis in which the reduced Fe magnon sector, the Tm Bloch equations, the joint EOM, and the detailed Liouville pathways are developed only after the Hamiltonian and 2DCS setup are in place (Secs. IX–XIV).
5. A Tm–Fe coupling sector (Sec. V) with bond-resolved linear exchange and on-site Van Vleck back-action, classified by the $Pbnm$ point group as $A_1^\pm$ and $A_2^\pm$, with the on-site inversion-pair cancellation analyzed explicitly.
6. A CPTP-compliant Tm dissipator (Sec. IX) and a Liouville-pathway derivation of every peak in the six-peak target list (Sec. XIII).
7. A coherent-emission expression with the SI impedance prefactor Z_0/c on the magnetic-dipole channel (Sec. XI).
8. A bond enumeration for the Fe–Fe and Fe–Tm shells (App. D) and a glossary of normalization conventions (App. A) so that every prefactor in every formula can be traced back to a single source.

The organizational logic of the manuscript is therefore: first derive the Hamiltonian only (Secs. II–VII); then introduce the 2DCS setup and experimental geometries (Sec. VIII); only after that do we develop the reduced dynamical equations, coherent emission, Liouville pathways, and perturbative EOM treatment (Secs. IX–XIV).

B. Notation

We use $\hbar = 1$ in all dynamical equations (restored in final numerical evaluations). μ_B is the Bohr magneton, $g_J = 7/6$ the Tm^{3+} Landé factor in 3H_6 , and $\gamma_e = g_e \mu_B / \hbar$ the Fe gyromagnetic ratio (with $g_e \approx 2$). The eight Gell-Mann matrices satisfy $\text{Tr}(\lambda^a \lambda^b) = 2\delta^{ab}$, $[\lambda^a, \lambda^b] = 2if^{abc}\lambda^c$, $\{\lambda^a, \lambda^b\} = \frac{4}{3}\delta^{ab} + 2d^{abc}\lambda^c$. Cartesian indices are $a, b, c \in \{x, y, z\}$ for spin and $\mu, \nu \in \{1, 2, 3, 4\}$ for the four Fe (or Tm) sublattices. The Bertaut basis variables are **F** (ferromagnetic), **G** (G-type AFM), **C** (C-type AFM), **A** (A-type AFM).

II. CRYSTAL STRUCTURE AND SUBLATTICE FRAMES

A. Pbnm setting and Wyckoff positions

TmFeO_3 adopts the orthorhombic $Pbnm$ setting (space group No. 62, historical orthoferrite convention; the IT standard $Pnma$ differs by an axis permutation). Lattice constants at 4 K are $a = 5.2534 \text{ \AA}$, $b = 5.5707 \text{ \AA}$, $c = 7.6076 \text{ \AA}$ [15]. The four Fe sublattices occupy Wyckoff position $4b$ ($\bar{1}$ site symmetry):

$$\begin{aligned} \text{Fe}_1 &= (0, \tfrac{1}{2}, \tfrac{1}{2}), & \text{Fe}_2 &= (\tfrac{1}{2}, 0, \tfrac{1}{2}), \\ \text{Fe}_3 &= (\tfrac{1}{2}, 0, 0), & \text{Fe}_4 &= (0, \tfrac{1}{2}, 0). \end{aligned} \quad (1)$$

The four Tm sublattices occupy Wyckoff position $4c$ (mirror m site symmetry, $z = \frac{1}{4}$ or $\frac{3}{4}$):

$$\begin{aligned} \text{Tm}_1 &= (0.0211, 0.9284, \tfrac{3}{4}), \\ \text{Tm}_2 &= (0.5211, 0.5716, \tfrac{1}{4}), \\ \text{Tm}_3 &= (0.4789, 0.4284, \tfrac{3}{4}), \\ \text{Tm}_4 &= (0.9789, 0.0716, \tfrac{1}{4}). \end{aligned} \quad (2)$$

The space-group generators in the $Pbnm$ setting are

$$S_1 = \{2_{010} \mid \tfrac{1}{2}\bar{1}0\}, \quad S_2 = \{2_{100} \mid \tfrac{1}{2}00\}, \quad I = \{\bar{1} \mid 000\}, \quad (3)$$

generating the D_{2h} point group $\{E, 2_x, 2_y, 2_z, I, m_x, m_y, m_z\}$ with the screw and glide translations as listed.

B. Sublattice permutations

Acting on the four Fe and four Tm sublattices, the generators permute as

$$\begin{aligned} \sigma_{S_1}^{\text{Fe}} &= (14)(23), & \sigma_{S_2}^{\text{Fe}} &= (12)(34), & \sigma_I^{\text{Fe}} &= e, \\ \sigma_{S_1}^{\text{Tm}} &= (14)(23), & \sigma_{S_2}^{\text{Tm}} &= (12)(34), & \sigma_I^{\text{Tm}} &= (14)(23). \end{aligned} \quad (4)$$

Inversion is non-trivial on Tm because $4c$ does not sit at an inversion center; it is trivial on Fe because $4b$ does.

C. Local sublattice frames

a. Definition. The *local frame* at sublattice μ is the orthonormal triad $\{\hat{e}_\mu^a\}_{a=x,y,z}$ obtained by rotating the global crystal triad $\{\hat{e}^a\} = \{\hat{a}, \hat{b}, \hat{c}\}$ via the proper-rotation part of the $Pbnm$ space-group element that maps site 1 to site μ :

$$\hat{e}_\mu^a = (R_\mu)^a_b \hat{e}^b, \quad R_\mu = \text{point-group part of } g_\mu, \quad (5)$$

where $g_\mu \in Pbnm$ is any operation with $g_\mu \mathbf{r}_1 \equiv \mathbf{r}_\mu \text{ mod lattice}$. Because the four sites in a Wyckoff orbit are related by the four cosets of the trivial site-symmetry stabilizer in $D_{2h}/\{E, I\} \simeq D_2$, exactly four such rotational parts arise:

$$\{R_\mu\}_{\mu=1}^4 = \{E, C_{2x}, C_{2y}, C_{2z}\}. \quad (6)$$

This is the Klein four-group of proper π -rotations about the three crystal axes, isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

b. Derivation for Fe (4b). The Fe^{3+} ions occupy site symmetry $\bar{1}$, so the stabilizer of Fe_1 is just $\{E, I\}$ and the four cosets of $\{E, I\}$ in D_{2h} are represented by $\{E, C_{2x}, C_{2y}, C_{2z}\}$. Denote by $g_\mu \in Pbnm$ a representative whose point-group part is the μ -th of these four rotations; we label the four sublattices by

$$\text{Fe}_\mu \equiv g_\mu \text{Fe}_1 \text{ (mod lattice)}, \quad \text{point-part}(g_\mu) = R_\mu, \quad (7)$$

which is the convention used in Eq. (1). As diagonal matrices in the crystal basis,

$$\begin{aligned} R_1 &= \text{diag}(+, +, +), & R_2 &= \text{diag}(+, -, -), \\ R_3 &= \text{diag}(-, +, -), & R_4 &= \text{diag}(-, -, +), \end{aligned} \quad (8)$$

where $\eta_\mu^a = (R_\mu)^a_a$ is the eigenvalue of S^a under the proper rotation that maps site $1 \rightarrow \mu$. The screw *translations* are absorbed in the position labels $\mathbf{r}_\mu$ and do not act on the spin.

c. Spin transformation. Spin is an axial vector. Under any *proper* rotation $R \in SO(3)$ both polar and axial vectors transform as $V^a \rightarrow R^a_b V^b$. Since each R_μ is proper ($\det R_\mu = +1$ for all μ), the spin at site μ in the *global* frame is related to the spin in the *local* frame by

$$S_\mu^a|_{\text{global}} = (R_\mu)^a_b \tilde{S}_\mu^b|_{\text{local}}, \quad (9)$$

i.e. a per-component sign flip of the chosen η_μ^a . The screw translations $\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$, $\frac{1}{2}\mathbf{a}$, etc., are absorbed into the position labels $\mathbf{r}_\mu$ and do not act on the spin.

d. Derivation for Tm (4c). The Tm^{3+} ions sit at site symmetry $.m$. (the mirror m_z at $z = \frac{1}{4}, \frac{3}{4}$). Its stabilizer is $\{E, m_z\} = \{E, I \cdot C_{2z}\}$, so the four cosets in $D_{2h}/\{E, m_z\}$ are $\{E, m_z\}, \{C_{2x}, m_y\}, \{C_{2y}, m_x\}, \{C_{2z}, I\}$. Choosing the proper-rotation representative in each coset gives *the same* Klein four-group Eq. (6). Concretely, the same generators map $\text{Tm}_1 \rightarrow \text{Tm}_\mu$ in the $4c$ orbit (up to the $z = \frac{1}{4} \leftrightarrow \frac{3}{4}$ shift absorbed in $\mathbf{r}_\mu$):

$$R_\mu^{\text{Tm}} = R_\mu^{\text{Fe}} \quad (\mu = 1, \dots, 4), \quad (10)$$

so the same matrices Eq. (8) act on the projected dipoles $\mathbf{J}_\mu^{\text{eff}}, \mathbf{d}_\mu^{\text{eff}}$ at every Tm site. The mirror m_z in the stabilizer acts trivially on the $\hat{z}$ -axis chosen as a principal quantization axis at the Tm site, so it imposes no further restriction on the local triad beyond fixing the gauge $\hat{e}_\mu^z = \pm \hat{c}$.

e. Group structure. The four matrices form the Klein four-group under multiplication,

$$\Pi_{\mu\nu} \equiv R_\mu R_\nu = \text{diag}(\eta_\mu^x \eta_\nu^x, \eta_\mu^y \eta_\nu^y, \eta_\mu^z \eta_\nu^z) = R_{\mu \cdot \nu}, \quad (11)$$

where $\mu \cdot \nu$ is the Klein product (each component multiplied mod 2). This product controls the sign pattern of the bond-projected Heisenberg exchange: a bond $\langle \text{Fe}_\mu, \text{Fe}_\nu \rangle$ contributes $J \mathbf{S}_\mu^{\text{loc}} \cdot \Pi_{\mu\nu} \mathbf{S}_\nu^{\text{loc}}$ in the *local* frame, which generates the four sign patterns $\sigma^{F,G,C,A}$ used in the Bertaut decomposition (Sec. IV C).

D. Polar versus axial vectors under R_μ

The spin $\mathbf{S}$ and the projected magnetic dipole $\mathbf{J}^{\text{eff}}$ are *axial* vectors; the projected electric dipole $\mathbf{d}^{\text{eff}}$ is *polar*. Under a proper rotation both transform identically (as R); under inversion the polar vector picks up an extra sign while the axial does not. The sublattice frames R_μ in Eq. (8) are improper $\det R_\mu = \pm 1$, and one must distinguish:

$$R_\mu^{\text{axial}} = \text{diag}(\eta_\mu^x, \eta_\mu^y, \eta_\mu^z), \quad R_\mu^{\text{polar}} = \det(R_\mu) R_\mu^{\text{axial}}. \quad (12)$$

For $R_2 = R_x(\pi)$ (a proper rotation, $\det = +1$), $R_\mu^{\text{polar}} = R_\mu^{\text{axial}}$. For $R_3 = R_x(\pi)R_z(\pi)$ ($\det = +1$ also, product of two proper rotations), the same holds. So in the present case $R_\mu^{\text{polar}} = R_\mu^{\text{axial}}$ for all four sublattices, and the distinction does not bite. We retain the bookkeeping for clarity in the emission formula (Sec. XI), where $\partial_t \mathbf{P}$ and $\partial_t \mathbf{M}$ enter together and the local-to-global rotation of $\mathbf{d}^{\text{eff}}$ versus $\mathbf{J}^{\text{eff}}$ must be applied with the right $R_\mu^{\text{polar/axial}}$.

E. Action on the projected qutrit subspace

A subtle point that affects the validity of the truncation to the lowest three Tm singlets: the rotation $R_x(\pi) = e^{-i\pi J_x}$ acts on a single $J = 6$ multiplet exactly via Wigner d -functions, $R_x(\pi)|J, m\rangle = (-1)^{J-m}|J, -m\rangle$, and therefore does *not* leak out of the $J = 6$ multiplet. The leakage out of the truncated three-level subspace $\{|E_1\rangle, |E_2\rangle, |E_3\rangle\}$ arises because the CEF parameters $B_l^{[m]}$ and $B_l^{-[m]}$ are both nonzero (Sec. III), breaking the $m \rightarrow -m$ symmetry as a separate good quantum number. Hence eigenstates $|E_\alpha\rangle$ do not have definite $m \rightarrow -m$ parity, and $R_x(\pi)|E_1\rangle$ is a linear combination of all even- m_J eigenstates including $|E_4\rangle, |E_5\rangle, \dots$ at energies $\gtrsim 14$ meV. Truncating to $\{E_1, E_2, E_3\}$ then makes

the projected matrix

$$[U_{R_x(\pi)}]_{\alpha\beta} \equiv \langle E_\alpha | e^{-i\pi J_x} | E_\beta \rangle \quad (\alpha, \beta \in \{1, 2, 3\}) \quad (13)$$

non-unitary by an amount $\|U^\dagger U - \mathbb{1}\|_{\text{op}} \approx 1.3$ in operator norm (numerically, with the CEF parameters of Sec. III); the missing weight sits in the discarded singlets.

For modeling purposes this means we should not use $R_x(\pi)$ as a unitary symmetry on the qutrit. Instead, we use R_μ only at the *vector* level, where it acts as a rotation on the projected dipoles $\mathbf{J}^{\text{eff}}, \mathbf{d}^{\text{eff}}$ themselves (3-vectors), not on the underlying Hilbert space. The qutrit Hilbert space has the same Gell-Mann basis on every sublattice and is sublattice-independent in our convention.

III. CRYSTAL-FIELD MODEL FOR TM³⁺

A. Stevens-operator Hamiltonian

a. Local-frame convention for the Tm sector. All single-ion operators introduced in this section — the Stevens operators $\hat{O}_l^m(J)$, the angular-momentum components J_x, J_y, J_z , the projected $\mathbf{J}^{\text{eff}}, \mathbf{d}^{\text{eff}}$, and the Gell-Mann λ^a in Sec. III C below — are defined *in the local triad* $\{\hat{e}_\mu^a\}$ of Sec. II C at a representative Tm site, taken to be $\mu = 1$. Because the four Tm sublattices are related by the proper rotations $R_\mu \in \{E, C_{2x}, C_{2y}, C_{2z}\}$ (Eq. (10)), the same Stevens parameters B_l^m diagonalize the same operator $H_{\text{CEF}}^{\text{Tm}}$ at every sublattice when one uses the local triad on that sublattice. The full four-sublattice CEF Hamiltonian is therefore

$$H_{\text{CEF}}^{\text{Tm}, 4-\text{sl}} = \sum_{\mu=1}^4 \sum_{l,m} B_l^m \hat{O}_l^m(\tilde{J}_\mu), \quad \tilde{J}_\mu^a = (R_\mu^{-1})^a_b J_\mu^b, \quad (14)$$

where $\tilde{J}_\mu^a$ is the local-frame component of the global angular momentum J_μ^b at site μ . Equivalently, in the local basis the spectrum, the eigenstates $|E_n\rangle$, and the matrix elements $M_{nn}^{(a)}$, tabulated in Eq. (98) are μ -independent; the μ -dependence enters only through R_μ when one re-expresses local-frame matrix elements as global tensor components — exactly the construction used in Sec. III F to assemble Bertaut combinations Λ_X^a .

In the orthorhombic $Pbnm$ environment, the TM^{3+} ion (ground multiplet 3H_6 , $J = 6$) sits at a site of C_s symmetry, with the mirror m_z perpendicular to the c axis at $z = \frac{1}{4}$. Acting on the $J = 6$ multiplet, $m_z = R_z(\pi) \cdot I = R_z(\pi)$ on axial $\mathbf{J}$ (since inversion is trivial on axial vectors), so $m_z|J, m_J\rangle = (-1)^{m_J}|J, m_J\rangle$. A Stevens operator $\hat{O}_l^m$ has matrix elements with $\Delta m_J = \pm|m|$, so $m_z \hat{O}_l^m m_z^{-1} = (-1)^{|m|} \hat{O}_l^m$. Invariance under the site mirror therefore requires $|m|$ *even*; both $\cos(m\phi)$ and $\sin(m\phi)$ partners are independently allowed because m_z does not relate $+|m|$ and $-|m|$ (this is the role of the additional C_2 axes of the full D_{2h} , which are absent here).

TABLE I. Stevens parameters B_l^m for Tm^{3+} in TmFeO_3 at low temperature, in meV, in the Hutchings convention with Stevens factors absorbed. These are the values that, fed into Eq. (15) and diagonalized on the full 13-dim $J = 6$ space, reproduce the cited $\epsilon_{2,3,4}$ and the wavefunction amplitudes of Tab. II verbatim (App. A).

B_2^0	-0.529	B_4^0	-1.3×10^{-4}	B_6^0	$+2.0 \times 10^{-6}$
B_2^2	-0.135	B_4^2	-1.7×10^{-3}	B_6^2	-1.1×10^{-5}
B_2^{-2}	+1.279	B_4^{-2}	$+3.29 \times 10^{-3}$	B_6^{-2}	-9.0×10^{-6}
		B_4^4	-1.22×10^{-3}	B_6^4	$+6.1 \times 10^{-5}$
		B_4^{-4}	-9.57×10^{-3}	B_6^{-4}	$+3.0 \times 10^{-6}$
				B_6^6	-9.0×10^{-6}
				B_6^{-6}	0

The CEF Hamiltonian therefore has the restricted form

$$H_{\text{CEF}}^{\text{Tm}} = \sum_{l=2,4,6} \sum_{\substack{m=-l \\ |m| \text{ even}}}^{+l} B_l^m \hat{O}_l^m(J), \quad (15)$$

where $\hat{O}_l^m$ are the Stevens operator equivalents in the convention of Hutchings [13], with the Stevens factors $\alpha_J = 1/99$, $\beta_J = 8/49\,005$, $\gamma_J = -5/891\,891$ for Tm^{3+} absorbed into the quoted B_l^m . The $O_l^{|m|}$ are the $\cos(m\phi)$ partners (real combinations of J_+^m and J_-^m) and $O_l^{-|m|}$ are the $\sin(m\phi)$ partners (imaginary combinations); explicit formulas are tabulated in [13], e.g., $O_2^0 = 3J_z^2 - J(J+1)$, $O_2^2 = \frac{1}{2}(J_+^2 + J_-^2)$, $O_2^{-2} = \frac{1}{2i}(J_+^2 - J_-^2)$. Note that the presence of the sin-partners $O_l^{-|m|}$ (with purely imaginary matrix elements in the $|J, m_J\rangle$ basis) makes $H_{\text{CEF}}^{\text{Tm}}$ complex Hermitian, *not* real symmetric, in this basis whenever any $B_l^{-|m|} \neq 0$; in particular the 4K parameters of Tab. I satisfy $B_2^{-2} \neq 0$. Eigenstates of Eq. (15) are therefore generically complex in the $|J, m_J\rangle$ basis, and the “real-wavefunction” gauge invoked in Sec. III C below should be understood as a T -eigenstate gauge (which is always achievable because $[H_{\text{CEF}}, T] = 0$, see Sec. III C), not as a literal reality condition on the expansion coefficients $c_{\alpha, m}$.

The fitted Stevens parameters from inelastic neutron spectroscopy [12] are reproduced in Table I. They are intended to reproduce the lowest four singlet energies $\epsilon_1 = 0$, $\epsilon_2 = 1.920$ meV, $\epsilon_3 = 7.844$ meV, $\epsilon_4 = 14.27$ meV. Direct diagonalization of Eq. (15) on the 13-dim $J = 6$ multiplet using Tab. I (Hutchings convention, Stevens factors absorbed; verified by the script in App. A) yields $\epsilon_{1\dots 6} - \epsilon_1 = (0, 1.920, 7.844, 14.269, 21.222, 33.661)$ meV, reproducing the three lowest singlets within rounding.

B. Three-singlet manifold and mirror parity

Diagonalizing Eq. (15) on the 13-dimensional $J = 6$ multiplet yields three nondegenerate singlets in the bot-

TABLE II. Wavefunction amplitudes for the lowest three Tm singlets (from diagonalizing Eq. (15), in the T -symmetric gauge of Sec. III C). $|E_1(A_1)\rangle$ and $|E_3(A_2)\rangle$ are symmetric in $m \rightarrow -m$ ($t_1 = +1$, $t_3 = -1$ with the $(-1)^m$ -sign rule); $|E_2(A_1)\rangle$ is antisymmetric ($t_2 = -1$), so $c_{2,-m} = -c_{2,m}$ and in particular $c_{2,0} \equiv 0$.

m_J	$c_{1,m}$	$c_{2,m}$	$c_{3,m}$
± 6	+0.580	+0.636	—
± 5	—	—	+0.484
± 4	+0.322	+0.286	—
± 3	—	—	+0.402
± 2	+0.212	+0.118	—
± 1	—	—	+0.323
0	+0.168	0 ^a	—

^a Forced to vanish by $t_2 = -1$ antisymmetry; any nonzero value is rounding noise.

tom 8 meV. The site symmetry C_s has only two irreducible representations, A_1 (mirror m_z even) and A_2 (mirror m_z odd). The mirror $m_z = R_z(\pi)$ acts on $|J, m_J\rangle$ as $(-1)^{m_J}$, so the unique selection rule is $m_J \bmod 2$: even- m_J states are A_1 and odd- m_J states are A_2 . Diagonalization gives

$$\begin{aligned} |E_1(A_1)\rangle &= \sum_{m \text{ even}} c_{1,m} |m\rangle, \\ |E_2(A_1)\rangle &= \sum_{m \text{ even}} c_{2,m} |m\rangle, \\ |E_3(A_2)\rangle &= \sum_{m \text{ odd}} c_{3,m} |m\rangle. \end{aligned} \quad (16)$$

The map $m \rightarrow -m$ is *not* an element of C_s (only m_z is, which gives $m \bmod 2$, not $m \rightarrow -m$); hence the two A_1 states $|E_{1,2}\rangle$ are general real linear combinations of $|m\rangle$ with even m , with no further parity constraint between $c_{\alpha, m}$ and $c_{\alpha, -m}$. The matrix element $\langle E_1 | J_z | E_2 \rangle \neq 0$ does not require any such discrete parity assignment between them; it only requires the two states to be linearly independent within the 7-dimensional even- m block. Likewise $\langle E_2 | J_x | E_3 \rangle \neq 0$ requires that $|E_2\rangle$ have non-vanishing amplitude on $|m\rangle$ for some m adjacent (in the $J_{\pm}$ ladder) to a non-vanishing amplitude of $|E_3\rangle$, which is generic.

The numerical wavefunction amplitudes, with the parity assignment of Eq. (16), are listed in Table II. The signs of $c_{2, \pm m}$ are opposite (antisymmetric) and we list only $c_{2, m > 0}$ with the convention $c_{2, -m} = -c_{2, m}$. In particular, antisymmetry forces the $m = 0$ amplitude $c_{2, 0} = -c_{2, 0} = 0$ identically; any nonzero entry there in the table below is to be read as numerical rounding noise from the diagonalizer in a non- T -symmetric gauge, not as a physical amplitude.

Normalization: $\sum_m |c_{\alpha, m}|^2 = 1$ within rounding for all three states (verified to 0.5% accuracy; deviations come from neglected $|c| < 0.02$ admixtures at $|m| = 4, 6$ and

from the ϵ_4 singlet contribution that is excluded from the truncation).

C. Projected angular-momentum matrix elements: T -symmetric gauge

For a non-Kramers ion ($J = 6$, integer) every state satisfies $T^2 = +1$, and H_{CEF} commutes with T because every Stevens operator of even rank $l = 2, 4, 6$ is built from products of two angular-momentum factors (each T -odd) and is therefore T -even. Even though H_{CEF} is generically complex Hermitian in the $|J, m\rangle$ basis when $B_l^{-|m|} \neq 0$, the commutation $[H_{\text{CEF}}, T] = 0$ guarantees that every non-degenerate eigenstate $|E_\alpha\rangle$ can be taken to be a T -eigenstate by a global phase rotation:

$$T|E_\alpha\rangle = t_\alpha|E_\alpha\rangle, \quad t_\alpha \in \{+1, -1\}. \quad (17)$$

In any gauge in which the expansion $|E_\alpha\rangle = \sum_m c_{\alpha,m}|m\rangle$ has $c_{\alpha,m} \in \mathbb{R}$ (which can always be arranged *after* the T -eigenstate phase fix is performed; the resulting reality of $c_{\alpha,m}$ is then a *consequence*, not an assumption), direct computation using $T|J, m\rangle = (-1)^{J-m}|J, -m\rangle$ gives

$$T|E_\alpha\rangle = \sum_m c_{\alpha,m}^* (-1)^{J-m} |-m\rangle = t_\alpha \sum_m c_{\alpha,m} |m\rangle, \quad (18)$$

whence $c_{\alpha,m} = t_\alpha (-1)^{J-m} c_{\alpha,-m}^*$. With real $c_{\alpha,m}$ and $J = 6$ this collapses to $c_{\alpha,m} = t_\alpha (-1)^m c_{\alpha,-m}$, so $t_\alpha = +1$ requires the m -parity rule $c_{\alpha,m} = (-1)^m c_{\alpha,-m}$ (symmetric for even m , antisymmetric for odd m), and $t_\alpha = -1$ flips both rules. For the model wavefunctions of Tab. II, the script in App. A verifies $t_1 = +1$, $t_2 = -1$, $t_3 = -1$: $|E_1\rangle$ has only even m and is $m \rightarrow -m$ symmetric (consistent with $t_1(-1)^{\text{even}} = +1$); $|E_2\rangle$ has only even m and is antisymmetric (consistent with $t_2(-1)^{\text{even}} = -1$); $|E_3\rangle$ has only odd m and is symmetric (consistent with $t_3(-1)^{\text{odd}} = +1$, i.e. $t_3 = -1$).

Because two of the three states are T -anti-invariant, the T -symmetric gauge in which all projected operators $\mathbf{J}_a^{\text{eff}}$ are purely imaginary requires multiplying *both* $|E_2\rangle$ and $|E_3\rangle$ by i :

$$|E_2\rangle \longrightarrow i|E_2\rangle, \quad |E_3\rangle \longrightarrow i|E_3\rangle, \quad |E_1\rangle \longrightarrow |E_1\rangle. \quad (19)$$

With this gauge fix every off-diagonal matrix element $\langle E_\alpha | J_a | E_\beta \rangle$ ($\alpha \neq \beta$, $a = x, y, z$) is purely imaginary (verified to 10^{-16} in App. A, Sec. 4).

In this T -symmetric gauge the projected operators are

$$\begin{aligned} J_z^{\text{eff}} &= +5.264 i |E_2\rangle\langle E_1| + \text{H.c.} = +5.264 \lambda^2, \\ J_x^{\text{eff}} &= +2.392 i |E_3\rangle\langle E_1| + 0.913 i |E_3\rangle\langle E_2| + \text{H.c.} \\ &= +2.392 \lambda^5 + 0.913 \lambda^7, \\ J_y^{\text{eff}} &= -2.787 i |E_3\rangle\langle E_1| + 0.466 i |E_3\rangle\langle E_2| + \text{H.c.} \\ &= -2.787 \lambda^5 + 0.466 \lambda^7, \end{aligned} \quad (20)$$

where the Gell-Mann matrices are in the ordered basis $(|E_1\rangle, |E_2\rangle, |E_3\rangle)$. The numerical $|\langle E_1 | J_z | E_2 \rangle|^2 =$

$5.264^2 = 27.71$ matches the value extracted from inelastic neutron scattering [12] to better than 1%.

D. Projected electric dipole

The projected electric dipole on the Tm site (from admixture of opposite-parity $5d/4f^{n+1}5d^{-1}$ states, treated phenomenologically [20]) has matrix elements with the same selection rules as J_x and J_z at the C_s site:

$$\begin{aligned} d_x^{\text{eff}} &\in \text{span}(\lambda^1), & d_z^{\text{eff}} &\in \text{span}(\lambda^4, \lambda^6), \\ d_y^{\text{eff}} &\in \text{span}(\lambda^1) \end{aligned} \quad (21)$$

(site mirror reverses sign of d_y).

Quantitative magnitudes from a Judd-Ofelt analysis of orthoferrite oscillator strengths are typically $\sim 10^{-2}$ debye [21]; we treat the coefficients p_x, p_{z4}, p_{z6} as phenomenological, constrained by the polarization-resolved 2DCS amplitudes.

E. Mirror classification of Gell-Mann generators

The site mirror $m_z = R_z(\pi)$ acts on the qutrit as $\text{diag}(+1, +1, -1)$ in the (E_1, E_2, E_3) ordering (because $E_{1,2}$ have even m_J and E_3 has odd m_J , and $R_z(\pi)|m\rangle = (-1)^m|m\rangle$). We adopt the standard $C_s \times \{T\}$ labeling convention: the Mulliken letter denotes the spatial (mirror) parity ($A_1 \equiv \text{mirror-even}$, $A_2 \equiv \text{mirror-odd}$), and the superscript $\pm$ denotes time-reversal parity ($+ \equiv T\text{-even}$, $- \equiv T\text{-odd}$). This is the convention used throughout Sec. V D and beyond for the Fe Bertaut bilinears $\mathcal{I}_{A_1^+}^{\text{Fe}}, \mathcal{I}_{A_2^+}^{\text{Fe}}$. Using this convention, the eight Gell-Mann generators classify as

A_1^+ (mirror-even, T -even):	$\lambda^1, \lambda^3, \lambda^8$,	(22)
A_1^- (mirror-even, T -odd):	λ^2 ,	
A_2^+ (mirror-odd, T -even):	λ^4, λ^6 ,	
A_2^- (mirror-odd, T -odd):	λ^5, λ^7 .	

(Earlier drafts of this manuscript contained a labeling typo in which λ^2 was written as “ A_2^+ ” and $\lambda^{4,6}$ as “ A_1^- ”; the assignments above are the corrected ones, internally consistent with the Fe-bilinear classification of Eq. (56) and Eq. (57).) This classification organizes every coupling in the Hamiltonian and every selection rule in the 2DCS analysis below. Direct verification: in the ordered (E_1, E_2, E_3) basis the diagonal mirror $U_{m_z} = \text{diag}(1, 1, -1)$ commutes with $\lambda^{1,2,3,8}$ (mirror-even) and anticommutes with $\lambda^{4,5,6,7}$ (mirror-odd), while complex conjugation in this basis fixes the imaginary Gell-Mann matrices $\lambda^{2,5,7}$ as T -odd and the real ones $\lambda^{1,3,4,6,8}$ as T -even (in the T -symmetric gauge of Sec. III C, where T acts as plain complex conjugation on the qutrit basis).

F. Bertaut algebra for the Tm Gell-Mann operators

The classification Eq. (22) is a *single-site* C_s classification under the local site mirror m_z . The full crystal symmetry $Pbnm$ further organizes the four-sublattice sums of λ^a into Bertaut combinations transforming as definite irreps of the global point group — exactly as the Fe spins are organized into $\mathbf{F}, \mathbf{G}, \mathbf{C}, \mathbf{A}$. Carrying out this step is essential because (i) it tells us *which* sublattice combination of λ^a is excited by a uniform field, and (ii) it tells us which combination couples to each Fe Bertaut mode in Eqs. (55)–(58). We do this here.

a. Convention. As stated in Sec. III, the qutrit Hilbert space at every Tm sublattice μ uses the *same* ordered basis $\{|E_1\rangle, |E_2\rangle, |E_3\rangle\}$ defined in the *local* frame at sublattice 1; the Gell-Mann operator λ^a then has the same 3×3 matrix at every sublattice. The sublattice dependence enters not in λ^a itself but in the *interpretation* of λ^a as a global-frame tensor component. Concretely, the projected operator that on sublattice 1 represents the local-frame component J_a^{loc} represents on sublattice μ the global-frame component $(R_\mu^{-1})_{ab} J_b^{\text{glob}}$ (equivalently, $J_a^{\text{loc}}|_\mu = (R_\mu)_{ab} J_b^{\text{glob}}$), because R_μ is the orthogonal map from local to global at sublattice μ .

b. σ -vectors of axial / polar generators. For the Klein-four sublattice frames $\{R_\mu\} = \{E, C_{2x}, C_{2y}, C_{2z}\}$ of Eq. (8), the diagonal of R_μ in the cubic axes gives

the sign carried by each axial component $a \in \{x, y, z\}$ at sublattice μ :

$$\begin{aligned} \eta^x &= (+, +, -, -) \equiv \sigma_C, \\ \eta_\mu^a &\equiv (R_\mu)_{aa}, \quad \eta^y = (+, -, +, -) \equiv \sigma_A, \\ \eta^z &= (+, -, -, +) \equiv \sigma_G, \end{aligned} \quad (23)$$

together with the trivial $\sigma_F = (+, +, +, +)$. (All R_μ are proper rotations, so polar and axial vectors carry the *same* σ -vectors here, cf. Sec. IID.) Eq. (23) recovers the standard Bertaut assignment for $Pbnm$ orthoferrites: the Γ_2 irrep contains F_x, C_y, G_z — exactly the three axial Bertaut variables whose σ -vector is $\sigma_C, \sigma_A, \sigma_G$ weighted by the local-axis label $\{x, y, z\}$.

c. σ -vectors of the eight Gell-Mann generators. Using the projected-operator decomposition Eqs. (98)–(21), each λ^a inherits the σ -vector of the local-frame Cartesian operator that it represents. The “represents (local)” entries below are the *inverse* of the matrix-element decomposition $J_x^{\text{eff}} = a\lambda^5 + b\lambda^7$, $J_y^{\text{eff}} = c\lambda^5 + d\lambda^7$ of Eq. (98): solving for $\lambda^{5,7}$ in terms of $J_{x,y}^{\text{eff}}$ gives $\lambda^5 = (dJ_x - bJ_y)/(ad - bc)$ and $\lambda^7 = (-cJ_x + aJ_y)/(ad - bc)$, with the common denominator $ad - bc \approx 3.66$ for $(a, b, c, d) = (2.392, 0.913, -2.787, 0.466)$. The entries below are written without that overall normalization, which does not affect the σ -vector decomposition (only the absolute scale).

λ^a	represents (local), $\propto$	σ -vector	couples (uniform field) to
λ^2	J_z	$\sigma_G = (+, -, -, +)$	G_z (Fe Néel)
λ^5	$dJ_x - bJ_y$	$d\sigma_C - b\sigma_A$	lin. comb of C_x, A_y
λ^7	$-cJ_x + aJ_y$	$-c\sigma_C + a\sigma_A$	lin. comb of C_x, A_y
λ^4	d_z (polar)	σ_G	G_z^{pol}
λ^6	d_z (polar)	σ_G	G_z^{pol}
λ^1	$d_{x,y}$ (polar)	lin. comb of σ_C, σ_A	electric drive
λ^3, λ^8	populations	σ_F	uniform n_a

with $(a, b, c, d) = (2.392, 0.913, -2.787, 0.466)$ from Eqs. (98).

d. Bertaut sublattice combinations. Define the four Tm-sublattice combinations of any λ^a in direct analogy with Eq. (36):

$$\Lambda_X^a = \frac{1}{4} \sum_{\mu=1}^4 \sigma_\mu^X \lambda_\mu^a, \quad X \in \{F, G, C, A\}. \quad (25)$$

Then a uniform external perturbation that couples to the global axial component J_a^{glob} acts only on the Bertaut

combination whose σ -vector matches η^a :

$$\sum_\mu (R_\mu \mathbf{h})_a (J_a^{\text{loc}})|_\mu = \sum_\mu \eta_\mu^a h_a (J_a^{\text{loc}})|_\mu \implies h_a \cdot 4\Lambda_{X(a)}^{[J_a^{\text{loc}}]}, \quad (26)$$

where $X(a) \in \{F, C, A, G\}$ is the irrep of η^a . In particular,

- Uniform h_z excites Λ_G^2 and (via $\lambda^{4,6}$ electric channel for e_z) $\Lambda_G^{4,6}$ — not Λ_F^2 . This is the precise meaning of the row $\lambda^2 \rightarrow G_z$ in Eq. (24).
- Uniform h_x excites the σ_C -component of $d\Lambda^5 - b\Lambda^7$, and uniform h_y the σ_A -component of $-c\Lambda^5 + a\Lambda^7$. The

four orthogonal Bertaut combinations $\Lambda_{C,A}^{5,7}$ are independent dynamical variables but only specific linear combinations are dipole-active under uniform drive.

- Populations $\lambda^{3,8}$ have σ_F , so only the uniform combination $\Lambda_F^{3,8}$ is generated by uniform pump (the homogeneous Tm population grating).

e. Polar (electric) drive — same R_μ , different parity. The polar generators λ^1 (d_x, d_y channel) and $\lambda^{4,6}$ (d_z channel) carry through the local-frame construction *identically* to the axial case, because the Klein-four sublattice rotations $R_\mu \in \{E, C_{2x}, C_{2y}, C_{2z}\}$ are all proper ($\det R_\mu = +1$, Sec. IID): polar and axial vectors share the same σ -vectors $\sigma_C, \sigma_A, \sigma_G$. Concretely, the projected polar-dipole operator obeys the same transport law as J_a ,

$$d_a^{\text{loc}}|_\mu = (R_\mu)_{ab} d_b^{\text{glob}}, \quad \eta_\mu^{a,\text{pol}} = \eta_\mu^{a,\text{axial}} = (R_\mu)_{aa}, \quad (27)$$

with no extra inversion sign because R_μ does not contain I (the inversion has been pushed into the site stabilizer / position labels in Sec. IIC). The electric-field analogue of Eq. (26) is therefore

$$-\sum_\mu E_a d_a^{\text{glob}}|_\mu = -E_a \sum_\mu \eta_\mu^a d_a^{\text{loc}}|_\mu \implies -E_a \cdot 4\Lambda_{X(a)}^{[d_a^{\text{loc}}]}, \quad (28)$$

with the same Klein-four irrep map $X(a) \in \{F, C, A, G\}$. Reading off the rows of Eq. (24) now from the polar side:

- Uniform E_z excites Λ_G^4 and Λ_G^6 (the G_z -staggered electric channel between $|E_{1,2}\rangle$ and $|E_3\rangle$). This is the electric-dipole counterpart of the Λ_G^2 Zeeman channel and shares the same σ_G pattern, so E_z and h_z probe the *same* Bertaut irrep — the distinguishing label is the matrix element ($\propto \mathbf{d}^{\text{eff}}$ for E_z vs. $\propto \mathbf{J}^{\text{eff}}$ for h_z), not the sublattice symmetry.
- Uniform E_x excites σ_C -projected Λ^1 , i.e. $E_x \cdot p_x \Lambda_C^1$, since the local-frame matrix element of d_x coincides

(up to the phenomenological coefficient p_x) with the λ^1 generator. Uniform E_y similarly excites $E_y \cdot p_y \Lambda_A^1$. Crucially the *same* λ^1 operator carries both: the orthogonal σ_C, σ_A combinations of the four λ_μ^1 are independent dynamical variables, and E_x, E_y select different ones. (This is the polar dual of the $\lambda^{5,7}$ statement above.)

- No uniform E component excites Λ_F^a for $a = 1, 4, 6$, because no $\eta_\mu^{a,\text{pol}} = + + + +$ exists in the Klein four-group — exactly as for axial drives.

The $|E_n\rangle$ basis-parity assignment from Sec. IIIB ($|E_1\rangle, |E_2\rangle \in A^+$ and $|E_3\rangle \in A^-$ under m_z) is what forces d_z to populate *only* λ^4, λ^6 (off-diagonal between opposite-parity states) and $d_{x,y}$ to populate *only* λ^1 (off-diagonal between same-parity states $|E_1\rangle \leftrightarrow |E_2\rangle$); the local-frame map R_μ then promotes this single-site selection rule to the four-sublattice $\sigma_{F,G,C,A}$ structure.

f. Why Fe has no polar dipole, but Tm does. The same local-frame analysis applied to the Fe sublattice would give $d_a^{\text{loc}}|_\mu = (R_\mu)_{ab} d_b^{\text{glob}}$ as well; however, the Fe site stabilizer $\{E, I\}$ at $4b$ contains *inversion*, which is polar-odd: $I d_a I^{-1} = -d_a$. The on-site projected electric dipole at Fe is therefore strictly zero by site symmetry, and only the rare-earth (Tm $4c$, stabilizer $\{E, m_z\}$, no inversion) carries an on-site electric dipole. This is why the electric channel of Eqs. (24) and (28) acts entirely through the Tm λ^a sector — there is no parallel “polar Bertaut amplitude” on Fe to populate. Magnetic dipole, by contrast, is even under I and therefore survives at both Fe ($4b$) and Tm ($4c$) sites, which is why the axial Bertaut variables $\mathbf{F}, \mathbf{G}, \mathbf{C}, \mathbf{A}$ exist on Fe and the parallel $\Lambda_X^{2,5,7}$ combinations exist on Tm.

g. Cross-coupling to Fe. The Van Vleck back-action Eq. (55), written explicitly with sublattice indices, becomes

$$H_{\text{aniso-Tm}} = -4 \sum_{\mathbf{n}} \left[u_1 \Lambda_{X_1}^1 \cdot \mathcal{I}_{X_1}^{\text{Fe}} + u_3 \Lambda_F^3 \mathcal{I}_{A_1^+}^{\text{Fe}} + u_8 \Lambda_F^8 \mathcal{I}_{A_1^+}^{\text{Fe}} + \dots \right], \quad (29)$$

i.e., for each λ^a only the Λ_X^a Bertaut combination whose σ -vector matches the irrep of the Fe bilinear $\mathcal{I}_X^{\text{Fe}}$ is allowed. In the Γ_2 phase the only Fe bilinears with nonzero expectation value are in A_1^+ ($G_z^2, F_x^2, C_y^2, \dots$, all σ_F -pattern), so the coupled Tm combinations are $\Lambda_F^{3,8}$ (population channel) and the σ_F -projection of Λ^1 (polar coherence channel). The $\bar{\chi}^{2z} G_z \lambda^2$ linear coupling of Eq. (58) acts on Λ_G^2 specifically (matching $\sigma_G \cdot \sigma_G = \sigma_F$).

h. Consequence for the scalar EOMs of Sec. XIV C. The amplitudes $\rho_{12}, \rho_{13}, \rho_{23}, n_1, n_2$ in Eqs. (118)–(122) should be read as *Bertaut-projected* amplitudes: $\rho_{12} \equiv$

Λ_G^2 (the G_z -staggered combination), and similarly ρ_{13}, ρ_{23} are the appropriate σ_C, σ_A linear combinations of $\Lambda^{5,7}$ that couple to uniform h_x, h_y . With this identification, the scalar EOMs are the correct Pbnm-allowed equations of motion at $\mathbf{q} = 0$: every amplitude carries a definite global irrep, every coupling vertex respects the four-sublattice σ -vector algebra, and the six-peak intrinsic count of Sec. IIIG is preserved.

IV. FE FOUR-SUBLATTICE HAMILTONIAN

A. Microscopic Hamiltonian

The Fe sector is the standard four-sublattice orthoferrite Hamiltonian with Heisenberg exchange (J_1 in-plane NN, J_{1c} out-of-plane NN, J_2 NNN), Dzyaloshinskii–Moriya (DM) interaction with two independent DM components D_1, D_2 allowed by $Pbnm$, and biaxial single-ion anisotropy with axes locked to $\hat{a}, \hat{b}, \hat{c}$:

$$\begin{aligned}
 H_{\text{Fe}} = & \sum_{\langle ij \rangle_{\text{NN}, \parallel}} J_1 \mathbf{S}_i \cdot \mathbf{S}_j + \sum_{\langle ij \rangle_{\text{NN}, c}} J_{1c} \mathbf{S}_i \cdot \mathbf{S}_j \\
 & + \sum_{\langle\langle ij \rangle\rangle_{\text{NNN}}} J_2 \mathbf{S}_i \cdot \mathbf{S}_j \\
 & + \sum_{\langle ij \rangle_{\text{NN}, \parallel}} \mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j) \\
 & - \sum_i [K_a (S_i^x)^2 + K_b (S_i^y)^2 + K_c (S_i^z)^2],
 \end{aligned} \tag{30}$$

with the standard hierarchy $J_1 \sim 4.5 \text{ meV} \gg |D_1|, |D_2| \sim 0.1 \text{ meV} \gg K_{a,b,c} \sim 10 \mu\text{eV}$. The DM bond pattern dictated by $Pbnm$ is [16, 17]:

$$\mathbf{D}_{ij}^{(z=\frac{1}{2})} = (0, +D_1, +D_2), \quad \mathbf{D}_{ij}^{(z=0)} = (0, -D_1, +D_2), \tag{31}$$

on in-plane NN bonds, with the partial sign reversal between $z = \frac{1}{2}$ and $z = 0$ planes required by the S_1 screw mapping the two planes (which flips $\hat{y}$ but preserves $\hat{z}$). Out-of-plane and NNN couplings have $\mathbf{D} = 0$ by inversion symmetry of the bond midpoint.

B. Local-frame form of H_{Fe}

For the Bertaut decomposition (Sec. IV C) and for the linearization around the Γ_2 ground state in Sec. XIV it is essential to recast Eq. (30) in the *local* sublattice frames defined by Eq. (8). Substituting $\mathbf{S}_\mu = R_\mu \tilde{\mathbf{S}}_\mu^{\text{loc}}$ (Eq. (9)) into each term gives:

a. Heisenberg term.

$$\mathbf{S}_\mu \cdot \mathbf{S}_\nu = \tilde{\mathbf{S}}_\mu^{\text{loc}} \cdot (R_\mu^\top R_\nu) \tilde{\mathbf{S}}_\nu^{\text{loc}} = \tilde{\mathbf{S}}_\mu^{\text{loc}} \cdot \Pi_{\mu\nu} \tilde{\mathbf{S}}_\nu^{\text{loc}}, \tag{32}$$

where $\Pi_{\mu\nu} = R_\mu R_\nu = \text{diag}(\eta_\mu^x \eta_\nu^x, \eta_\mu^y \eta_\nu^y, \eta_\mu^z \eta_\nu^z)$ (Eq. (11); we used $R_\mu^\top = R_\mu^{-1} = R_\mu$ for involutions). The diagonal sign pattern of $\Pi_{\mu\nu}$ is exactly the per-component selection rule that produces the Bertaut sign vectors $\sigma_\mu^{F,G,C,A} \sigma_\nu^{F,G,C,A}$ on quadratic mode combinations: a NN bond $\langle \mu\nu \rangle$ between sublattices in the same z -plane has $\Pi_{\mu\nu} = \text{diag}(+, -, -)$ for the in-plane pair $(\mu, \nu) = (1, 4)$ etc., enforcing the antiferromagnetic G -staggering of S^z and the A/C -pattern of $S^{x,y}$ at the classical Γ_2 minimum (uniform $\tilde{S}_\mu^z = +S$ in every local frame).

b. DM term. Under proper rotations the cross product transforms as $R(\mathbf{u} \times \mathbf{v}) = R\mathbf{u} \times R\mathbf{v}$, so

$$\mathbf{D}_{\mu\nu} \cdot (\mathbf{S}_\mu \times \mathbf{S}_\nu) = \tilde{\mathbf{D}}_{\mu\nu}^{\text{loc}} \cdot (\tilde{\mathbf{S}}_\mu^{\text{loc}} \times \Pi_{\mu\nu} \tilde{\mathbf{S}}_\nu^{\text{loc}}), \tag{33}$$

with $\tilde{\mathbf{D}}_{\mu\nu}^{\text{loc}} = R_\mu^{-1} \mathbf{D}_{\mu\nu}$. The sign reversal of D_1 between the $z = 0$ and $z = \frac{1}{2}$ planes in Eq. (31) is the nontrivial part of the $Pbnm$ pattern that survives in the local coordinates, but it does *not* imply a standalone $F_x G_z$ Lifshitz invariant in the Bertaut convention of Eq. (37). Direct bond-sum projection of the written DM Hamiltonian gives the channel structure quoted in Eq. (46): D_1 mixes the (A, F) and (G, C) sectors, whereas D_2 mixes the (A, C) and (F, G) sectors.

c. Anisotropy. Because $K_{a,b,c}$ are diagonal in the *global* cubic basis and R_μ permutes signs but not axes,

$$\sum_a K_a (S_\mu^a)^2 = \sum_a K_a [(R_\mu)^a_b \tilde{S}_\mu^b]^2 = \sum_a K_a (\tilde{S}_\mu^a)^2, \tag{34}$$

i.e. the single-ion anisotropy is μ -independent and *uniform* in the local frame. This is the formal reason every Fe sublattice has its easy axis along the same local $\hat{e}_\mu^z$ (the global $\hat{c}$ -axis times η_μ^z), so the Γ_2 phase has uniform $\tilde{S}_\mu^z = +S$ in local frames.

d. Local-frame Hamiltonian. Combining Eqs. (32)–(34) the Fe Hamiltonian Eq. (30) reads, in local frames,

$$\begin{aligned}
 \tilde{H}_{\text{Fe}} = & \sum_{\langle \mu\nu \rangle} J_{\mu\nu} \tilde{\mathbf{S}}_\mu^{\text{loc}} \cdot \Pi_{\mu\nu} \tilde{\mathbf{S}}_\nu^{\text{loc}} \\
 & + \sum_{\langle \mu\nu \rangle} \tilde{\mathbf{D}}_{\mu\nu}^{\text{loc}} \cdot (\tilde{\mathbf{S}}_\mu^{\text{loc}} \times \Pi_{\mu\nu} \tilde{\mathbf{S}}_\nu^{\text{loc}}) \\
 & - \sum_\mu [K_a (\tilde{S}_\mu^x)^2 + K_b (\tilde{S}_\mu^y)^2 + K_c (\tilde{S}_\mu^z)^2],
 \end{aligned} \tag{35}$$

in which the exchange-dominated reference configuration is the uniform local state $\tilde{\mathbf{S}}_\mu^{\text{loc}} = +S\hat{e}^z$ for all μ . The weak cantings that generate the orthoferrite order parameters then appear as transverse, generally non-uniform fluctuations about this reference state. Linearizing about it (Holstein–Primakoff or **L**-and-**M** method) is equivalent to the linearization of the Bertaut variables **F**, **G**, **C**, **A** used in Sec. IV C, with the Klein-product $\Pi_{\mu\nu}$ playing the role of the bond-projector of Eq. (39): the Fourier transform on the four-site Klein index diagonalizes Eq. (35) into the four $X \in \{F, G, C, A\}$ blocks.

C. Bertaut decomposition

Define the four Bertaut sublattice combinations

$$\begin{aligned}
 \mathbf{F}_\mathbf{n} &= \frac{1}{4} \sum_\mu \mathbf{S}_{\mathbf{n},\mu}, & \mathbf{G}_\mathbf{n} &= \frac{1}{4} \sum_\mu \sigma_\mu^G \mathbf{S}_{\mathbf{n},\mu}, \\
 \mathbf{C}_\mathbf{n} &= \frac{1}{4} \sum_\mu \sigma_\mu^C \mathbf{S}_{\mathbf{n},\mu}, & \mathbf{A}_\mathbf{n} &= \frac{1}{4} \sum_\mu \sigma_\mu^A \mathbf{S}_{\mathbf{n},\mu},
 \end{aligned} \tag{36}$$

with sign vectors

$$\begin{aligned}\sigma^F &= (+, +, +, +), & \sigma^G &= (+, -, -, +), \\ \sigma^C &= (+, +, -, -), & \sigma^A &= (+, -, +, -).\end{aligned}\quad (37)$$

These satisfy the Klein four-group multiplication $\sigma_\mu^X \sigma_\mu^Y = \sigma_\mu^{XY}$ with the table

$$\begin{array}{c|cccc} \cdot & F & G & C & A \\ \hline F & F & G & C & A \\ G & G & F & A & C \\ C & C & A & F & G \\ A & A & C & G & F \end{array}\quad (38)$$

and the orthogonality $\sum_\mu \sigma_\mu^X \sigma_\mu^Y = 4\delta^{XY}$.

The inverse of Eq. (36) is

$$\mathbf{S}_{\mathbf{n},\mu} = \sum_X \sigma_\mu^X \mathbf{X}_{\mathbf{n}} \quad \text{for } X \in \{F, G, C, A\}, \quad (39)$$

i.e., $\mathbf{X}$ are mode amplitudes in units of *average sublattice spin*, not summed spin: $|\mathbf{F}|^2 + |\mathbf{G}|^2 + |\mathbf{C}|^2 + |\mathbf{A}|^2 = S^2$ at the spin-length constraint $|\mathbf{S}_\mu| = S$.

D. Effective fields and the per-cell normalization

We adopt one normalization throughout the manuscript and propagate it consistently: the effective field on a single Fe sublattice spin is

$$\mathbf{H}_{\mathbf{n},\mu}^{\text{eff}} = -\frac{\partial H_{\text{Fe}}}{\partial \mathbf{S}_{\mathbf{n},\mu}} \quad (\text{per-site definition}), \quad (40)$$

which has units of energy (energy per spin component). The Bertaut projection of the LLG equation gives, after substituting Eq. (39) and using the orthogonality $\sum_\mu \sigma_\mu^X \sigma_\mu^Y = 4\delta^{XY}$,

$$\begin{aligned}\dot{\mathbf{X}}_{\mathbf{n}} &= -\frac{1}{\hbar} \sum_{Y,Z: Y \cdot Z = X} \mathbf{Y}_{\mathbf{n}} \times \mathbf{H}_{\mathbf{n}}^{Z,\text{site}}, \\ \mathbf{H}_{\mathbf{n}}^{Z,\text{site}} &\equiv \frac{1}{4} \sum_\mu \sigma_\mu^Z \mathbf{H}_{\mathbf{n},\mu}^{\text{eff}},\end{aligned}\quad (41)$$

where the Klein-product structure $Y \cdot Z = X$ ensures only the allowed mode mixings appear. The auxiliary field $\mathbf{H}_{\mathbf{n}}^{Z,\text{site}}$ is the average per-site effective field weighted by σ_μ^Z . With this convention the precession prefactor is $1/\hbar$ (and equals unity in the $\hbar = 1$ dynamical equations used elsewhere in the manuscript), not $\gamma_e = g_e \mu_B / \hbar$, which would apply only if $\mathbf{H}^{\text{eff}}$ were a magnetic field rather than an energy derivative. There is likewise no extra prefactor in front of the Heisenberg or anisotropy contributions:

$$\mathbf{H}_{\text{aniso}}^{F,\text{site}} = +2\mathbf{K}\mathbf{F}, \quad \mathbf{H}_{\text{aniso}}^{G,\text{site}} = +2\mathbf{K}\mathbf{G}, \quad \text{etc.}, \quad (42)$$

where $\mathbf{K} = \text{diag}(K_a, K_b, K_c)$. *Derivation:* per site, $-\partial[-K_a(S_\mu^x)^2]/\partial S_\mu^x = +2K_a S_\mu^x$, so $\mathbf{H}_{\mathbf{n},\mu}^{\text{eff,aniso}} = 2\mathbf{K}\mathbf{S}_{\mathbf{n},\mu}$. Substituting $\mathbf{S}_\mu = \sum_X \sigma_\mu^X \mathbf{X}$ and projecting via Eq. (41),

$$\begin{aligned}\mathbf{H}_{\text{aniso}}^{Z,\text{site}} &= \frac{1}{4} \sum_\mu \sigma_\mu^Z \cdot 2\mathbf{K} \sum_Y \sigma_\mu^Y \mathbf{Y} \\ &= \frac{1}{4} \cdot 2\mathbf{K} \cdot 4\mathbf{Z} = 2\mathbf{K}\mathbf{Z},\end{aligned}\quad (43)$$

matching Eq. (42) cleanly.

Similarly the Heisenberg contribution gives, per Klein channel,

$$\begin{aligned}\mathbf{H}_{\text{Heis}}^{F,\text{site}} &= -2J^F \mathbf{F}, & \mathbf{H}_{\text{Heis}}^{G,\text{site}} &= -2J^G \mathbf{G}, \\ \mathbf{H}_{\text{Heis}}^{C,\text{site}} &= -2J^C \mathbf{C}, & \mathbf{H}_{\text{Heis}}^{A,\text{site}} &= -2J^A \mathbf{A},\end{aligned}\quad (44)$$

with the Bertaut-projected exchange constants

$$\begin{aligned}J^F &= +(z_1 J_1 + z_{1c} J_{1c} + z_2 J_2), \\ J^G &= -(z_1 J_1 + z_{1c} J_{1c} - z_2 J_2), \quad \text{etc.},\end{aligned}\quad (45)$$

where z_1, z_{1c}, z_2 are coordination numbers (Sec. C) and the relative signs follow from σ^X on the bond endpoints.

E. DM Hamiltonian in Bertaut form

We derive the DM Bertaut Lifshitz invariants by direct bond-sum projection. The DM term in Eq. (30) is a sum over in-plane NN bonds. Using the directed bond list implicit in Eq. (31) (four $\text{Fe}_1 \rightarrow \text{Fe}_2$ bonds in the $z = \frac{1}{2}$ plane with $\mathbf{D} = (0, +D_1, +D_2)$ and four $\text{Fe}_3 \rightarrow \text{Fe}_4$ bonds in the $z = 0$ plane with $\mathbf{D} = (0, -D_1, +D_2)$) and the Bertaut convention of Eq. (37), the $\mathbf{q} = 0$ reduction of the written bond Hamiltonian is (App. B)

$$H_{\text{DM}}^{\mathbf{q}=0} = N_c \left[-8D_1 ((\mathbf{F} \times \mathbf{G})_y + (\mathbf{C} \times \mathbf{A})_y) + 8D_2 ((\mathbf{G} \times \mathbf{C})_z - (\mathbf{F} \times \mathbf{A})_z) \right], \quad (46)$$

where N_c is the number of unit cells. The D_1 invariant $(\mathbf{F} \times \mathbf{G})_y = F_z G_x - F_x G_z$ is the standard Treves-Bertaut weak-ferromagnetic Lifshitz invariant of $Pbnm$ orthoferrites [16–18]: evaluated on the Γ_2 background it reduces to $-F_x G_z$, so the DM energy at the background is $+8N_c D_1 F_x G_z$. The D_2 invariants $(\mathbf{G} \times \mathbf{C})_z = G_x C_y - G_y C_x$ and $(\mathbf{F} \times \mathbf{A})_z = F_x A_y - F_y A_x$ are the corresponding mixing terms between the Γ_2 partner modes (G_z, C_y) and the Γ_4 excited modes. The overall sign is tied to the directed-bond convention; reversing every bond orientation is equivalent to $D_{1,2} \rightarrow -D_{1,2}$ and leaves the physics unchanged.

The DM contribution to the Bertaut effective fields $\mathbf{H}_{\text{DM}}^{X,\text{site}} = -\frac{1}{4} \partial(H_{\text{DM}}/N_c)/\partial \mathbf{X}$ is then

$$\begin{aligned}\mathbf{H}_{\text{DM}}^{F,\text{site}} &= (2D_1 G_z - 2D_2 A_y, -2D_2 A_x, 2D_1 G_x), \\ \mathbf{H}_{\text{DM}}^{G,\text{site}} &= (2D_1 F_z - 2D_2 C_y, 2D_2 C_x, -2D_1 F_x), \\ \mathbf{H}_{\text{DM}}^{C,\text{site}} &= (-2D_1 A_z + 2D_2 G_y, -2D_2 G_x, 2D_1 A_x), \\ \mathbf{H}_{\text{DM}}^{A,\text{site}} &= (2D_1 C_z - 2D_2 F_y, 2D_2 F_x, -2D_1 C_x).\end{aligned}\quad (47)$$

F. Ground state in the Γ_2 phase

The Γ_2 phase is defined by the order parameter pattern

$$\mathbf{G} = G_z \hat{z}, \quad \mathbf{F} = F_x \hat{x}, \quad \mathbf{C} = C_y \hat{y}, \quad \mathbf{A} \approx 0, \quad (48)$$

as the phenomenological identification of the ordered phase. However, for the Hamiltonian written in Eqs. (30) and (46), the DM sector does *not* generate a direct $D_1 F_x G_z$ or $D_2 C_y F_x$ energy inside the reduced $(\mathbf{F}, \mathbf{G}, \mathbf{C}, \mathbf{A})$ manifold. Instead it couples the nominal Γ_2 variables to the exchange-stiff partner modes A_z, C_x, G_y , and A_x , so a consistent static reduction requires either integrating out those auxiliary modes or adopting a different Bertaut-label convention. Accordingly we replace the naive canting estimate by the weaker, but correct, statement

$F_x/G_z, C_y/G_z$ must be obtained from a full static minimization of H_{Fe} (App. C).

(or from an explicit elimination of A_z, C_x, G_y, A_x .)

not from the truncated ansatz $F_x \sim D_1 G_z / H_E^{\text{exch}}$ alone. (49)

V. TM-FE COUPLING

A. Most general bilinear coupling

Before imposing time reversal, the most general bilinear Tm-Fe coupling allowed by $Pbnm$ would be a sum over all eight qutrit generators. Because the Hamiltonian itself must be T -even while the Fe spin is T -odd, only the T -odd Tm generators $\lambda^2, \lambda^5, \lambda^7$ [Eq. (22)] can appear linearly. The most general bilinear Tm-Fe coupling consistent with both $Pbnm$ and time reversal is therefore

$$H_{\text{Tm-Fe}}^{\text{lin}} = \sum_{\mathbf{n}, \mathbf{n}'} \sum_{j, i} \sum_{a \in \{2, 5, 7\}} \sum_{b=x, y, z} \chi_{ji}^{ab}(\mathbf{n} - \mathbf{n}') \lambda_{\mathbf{n}, j}^a S_{\mathbf{n}', i}^b, \quad (50)$$

where $\lambda_{\mathbf{n}, j}^a$ is the a th Gell-Mann generator on Tm sublattice j in cell $\mathbf{n}$, $S_{\mathbf{n}', i}^b$ is the b th spin component on Fe sublattice i in cell $\mathbf{n}'$, and $\chi_{ji}^{ab}(\mathbf{R})$ is a bond-resolved coupling tensor. The space group $Pbnm$ constrains χ via the symmetry-orbit relations of Sec. C; time reversal excludes any coefficient that would multiply the T -even generators $\lambda^{1, 3, 4, 6, 8}$ linearly.

For the on-site $\mathbf{q} = 0$ projection that controls the Bertaut-mode coupling, the relevant combinations are

$$\bar{\chi}^{ab} = \sum_{\mathbf{R}, j, i} \chi_{ji}^{ab}(\mathbf{R}), \quad (51)$$

i.e., the uniform sum over all bond shells. Because Tm sits at a non-centrosymmetric site (4c), and the inversion image of the bond from Tm _{j} to Fe _{i} at offset $\mathbf{R}$ goes from Tm_{(14)(23) j} to Fe _{i} at offset $-\mathbf{R}$. Because the Fe spin is an axial vector, it is *unchanged* by inversion; the inversion-pair cancellation at $\mathbf{q} = 0$ therefore acts only through the

mirror parity of the Tm operator, exactly as in App. C. The inversion-pair sums in Eq. (51) cancel several entries of $\bar{\chi}^{ab}$.

B. Surviving $\mathbf{q} = 0$ channels

After time reversal and the inversion-pair cancellation, the surviving $\mathbf{q} = 0$ linear couplings are:

- $\bar{\chi}^{2z} \neq 0$: couples λ^2 (A_1^- , mirror-even, T -odd; cf. Eq. (22)) to G_z (the Γ_2 Néel order parameter). *This is the dominant linear Fe-Tm exchange vertex used in Geometries II and III.*
- All off-diagonal mirror-odd couplings $\bar{\chi}^{5x}, \bar{\chi}^{7x}, \bar{\chi}^{5y}, \bar{\chi}^{7y}$ cancel on the $\mathbf{q} = 0$ projection due to the bond-pair structure of H_{Fe} (App. C).
- The T -even channels $\lambda^{1, 3, 4, 6, 8}$ therefore do not appear in the microscopic linear exchange at $\mathbf{q} = 0$; they enter only through the second-order macroscopic back-action discussed in Sec. VD.

This is the key statement that motivates the literature-aligned “anisotropy-quench” route (Sec. VD) for the apparent mirror-odd-to- F coupling: it does not exist as a $\mathbf{q} = 0$ linear exchange, but it does emerge at second order via the symmetric $(\lambda^1, \lambda^3, \lambda^8)$ Tm density coordinates.

C. Bare CEF field on the Tm qutrit

The on-site Tm Hamiltonian in the truncated three-level basis is $H_{\text{Tm}}^{(0)} = \sum_{\alpha} \epsilon_{\alpha} |E_{\alpha}\rangle \langle E_{\alpha}|$, which we expand in Gell-Mann basis as

$$H_{\text{Tm}}^{(0)} = \bar{\epsilon} \mathbf{1} + h_3^{(0)} \lambda^3 + h_8^{(0)} \lambda^8, \quad (52)$$

with

$$\begin{aligned} \bar{\epsilon} &= (\epsilon_1 + \epsilon_2 + \epsilon_3)/3, \\ h_3^{(0)} &= (\epsilon_1 - \epsilon_2)/2, \\ h_8^{(0)} &= (\epsilon_1 + \epsilon_2 - 2\epsilon_3)/(2\sqrt{3}). \end{aligned} \quad (53)$$

Numerically, $h_3^{(0)} = -0.96$ meV, $h_8^{(0)} = -3.97$ meV.

The *first-order static energy shift* from $\bar{\chi}^{2z} G_z \lambda^2$ vanishes in the Γ_2 ground state because $\langle \lambda^2 \rangle_0 = 0$ (the ground state $|E_1\rangle$ is real in the T -symmetric gauge). The operator itself is of course off-diagonal in the qutrit basis and mixes $|E_1\rangle$ with $|E_2\rangle$, so it contributes to static renormalization only from second order onward. There is no time-reversal-allowed linear $G_z \lambda^{3, 8}$ term in $H_{\text{Tm-Fe}}^{\text{lin}}$. Any static renormalization of the ϵ_{α} from the Fe background is therefore absorbed into the effective CEF parameters or into the second-order Van Vleck functional, not written as a first-order diagonal exchange term.

D. Anisotropy modulation: Van Vleck back-action

The dominant Tm-induced contribution to the Fe anisotropy at $\mathbf{q} = 0$ comes from second-order perturbation in the linear Tm-Fe coupling. The object introduced below is not the most general microscopic Tm-Fe Hamiltonian; it is the most general $\mathbf{q} = 0$ macroscopic effective back-action operator on the *retained* qutrit sector, to quadratic order in the Fe Bertaut amplitudes, generated by second-order virtual processes in the microscopic Tm-Fe coupling. For the Tm ground state, the same mechanism reduces to the familiar Van Vleck energy correction. At fixed $\mathbf{G} = G_z \hat{z}$, $\mathbf{F} = F_x \hat{x}$ is

$$\delta E_1^{\text{VV}} = - \sum_{\alpha \neq 1} \frac{|\langle E_\alpha | H_{\text{Tm-Fe}}^{\text{lin}} | E_1 \rangle|^2}{\epsilon_\alpha - \epsilon_1}, \quad (54)$$

which depends on G_z, F_x through the projected operators in $H_{\text{Tm-Fe}}^{\text{lin}}$. This remains consistent even though $H_{\text{Tm-Fe}}^{\text{lin}}$ contains only the T -odd generators $\lambda^{2,5,7}$ linearly: at second order, products of two T -odd operators are T -even, and the $\text{SU}(3)$ algebra closes onto the even sector, with $\{\lambda^2, \lambda^5\} = \lambda^6$, $\{\lambda^2, \lambda^7\} = -\lambda^4$, $\{\lambda^5, \lambda^7\} = \lambda^1$, while $(\lambda^2)^2, (\lambda^5)^2, (\lambda^7)^2$ contain the diagonal generators $\lambda^{3,8}$ (plus the identity). Expanding to quadratic order in the Bertaut amplitudes and decomposing in the mirror sector classification of Eq. (22), the resulting $\mathbf{q} = 0$ macroscopic back-action functional is

$$\begin{aligned} H_{\text{aniso-Tm}} = & - \sum_{\mathbf{n},j} \left[u_{1,j} \lambda_{\mathbf{n},j}^1 + u_{3,j} \lambda_{\mathbf{n},j}^3 \right. \\ & \left. + u_{8,j} \lambda_{\mathbf{n},j}^8 \right] \mathcal{I}_{A_1^+}^{\text{Fe}}(\mathbf{n}) \\ & - \sum_{\mathbf{n},j} \left[v_{4,j} \lambda_{\mathbf{n},j}^4 + v_{6,j} \lambda_{\mathbf{n},j}^6 \right] \mathcal{I}_{A_2^+}^{\text{Fe}}(\mathbf{n}), \end{aligned} \quad (55)$$

where the most general A_1^+ (mirror-even) Fe bilinears are

$$\mathcal{I}_{A_1^+}^{\text{Fe}} = \alpha_1 G_z^2 + \alpha_2 F_x^2 + \alpha_3 C_y^2 + \alpha_4 (G_x^2 - G_y^2) + \alpha_5 F_x C_y + \dots \quad (56)$$

and the A_2^+ (mirror-odd) ones are

$$\mathcal{I}_{A_2^+}^{\text{Fe}} = \beta_1 F_x G_z + \beta_2 G_z C_y + \dots \quad (57)$$

with α_i, β_i phenomenological $\mathbf{q} = 0$ coefficients inherited from the microscopic χ_{ji}^{ab} tensor and the Tm CEF matrix elements. We keep at least the two leading combinations $\alpha_1 G_z^2$ and $\alpha_2 F_x^2$ as independent; collapsing $\mathcal{I}_{A_1^+}^{\text{Fe}}$ to a single combination such as $G_z^2 - F_x^2$ would be a one-parameter truncation, not a symmetry constraint.

The A_2^+ on-site coupling $v_4 \lambda^4 F_x G_z$ vanishes in the inversion-pair cancellation analysis (App. C) at $\mathbf{q} = 0$,

leaving the A_1^+ Van Vleck channels as the dominant back-action mechanism.

E. Coherence contribution to Van Vleck

The Van Vleck back-action Eq. (55) carries both *population* and *coherence* contributions in the $\text{SU}(3)$ Bloch language. Writing $\rho_{\text{Tm}} = \sum_a \rho_a \lambda^a / 2 + \mathbb{1}/3$ and extracting the coefficient of each A_1^+ generator,

$$\begin{aligned} u_{1,j} \langle \lambda_{\mathbf{n},j}^1 \rangle & \sim \text{Re}[\rho_{12}^{(j)}(\mathbf{n})], \\ u_{3,j} \langle \lambda_{\mathbf{n},j}^3 \rangle & \sim p_1^{(j)}(\mathbf{n}) - p_2^{(j)}(\mathbf{n}), \\ u_{8,j} \langle \lambda_{\mathbf{n},j}^8 \rangle & \sim \frac{1}{\sqrt{3}}(p_1 + p_2 - 2p_3). \end{aligned} \quad (58)$$

The coherence term $\text{Re}[\rho_{12}(t)]$ oscillates at ω_{12} during the pump-delay coordinate and therefore modulates the Fe anisotropy through the Van Vleck channel in ideal Geometry I of the low-temperature Γ_2 phase.

VI. DRIVE HAMILTONIAN

The single-cycle THz pulse couples to both Fe and Tm sectors. The Fe Zeeman vertex (drives the qFM and qAFM modes through their magnetic-dipole matrix elements) and the Tm magnetic-dipole vertex (drives the $\omega_{12,13,23}$ CF coherences through $\mathbf{J}^{\text{eff}}$):

$$H_{\text{drive}}^{\text{m}}(t) = -g_e \mu_B \sum_{\mathbf{n},i} \mathbf{h}(t) \cdot \mathbf{S}_{\mathbf{n},i} - g_J \mu_B \sum_{\mathbf{n},j} \mathbf{h}(t) \cdot \mathbf{J}_{\mathbf{n},j}^{\text{eff}}. \quad (59)$$

The Tm electric-dipole vertex (drives the same CF coherences through $\mathbf{d}^{\text{eff}}$):

$$H_{\text{drive}}^{\text{e}}(t) = - \sum_{\mathbf{n},j} \mathbf{e}(t) \cdot \mathbf{d}_{\mathbf{n},j}^{\text{eff}}. \quad (60)$$

For THz propagation along $\hat{b}$, the two relevant field assignments are

$$\begin{aligned} H \parallel a, E \parallel c : & \quad h_x(t) \neq 0, e_z(t) \neq 0 \\ & \implies \text{couples } \lambda^{5,7} \text{ and } \lambda^{4,6}; \\ H \parallel c, E \parallel a : & \quad h_z(t) \neq 0, e_x(t) \neq 0 \\ & \implies \text{couples } \lambda^2 \text{ and } \lambda^1. \end{aligned} \quad (61)$$

A. Drive Hamiltonian in Bertaut form

Substituting the local-frame transport laws Eq. (9) (spin/ $\mathbf{J}^{\text{eff}}$, axial) and Eq. (27) ($\mathbf{d}^{\text{eff}}$, polar) into Eqs. (59)–(60) converts each four-sublattice sum into the Bertaut combination $\Lambda_{X(a)}^a$ matched to the σ -vector of the global field component (Eqs. (26) and (28)):

$$\begin{aligned}
H_{\text{drive}}^{\text{m,Tm}}(t) &= -4g_J\mu_B \sum_{\mathbf{n}} \left[h_z M_{12}^{(z)} \Lambda_G^2 + h_x (d \Lambda_C^5 - b \Lambda_C^7) + h_y (-c \Lambda_A^5 + a \Lambda_A^7) \right], \\
H_{\text{drive}}^{\text{e,Tm}}(t) &= -4 \sum_{\mathbf{n}} \left[e_z (p_{z4} \Lambda_G^4 + p_{z6} \Lambda_G^6) + e_x p_x \Lambda_C^1 + e_y p_y \Lambda_A^1 \right], \\
H_{\text{drive}}^{\text{m,Fe}}(t) &= -4g_e\mu_B \sum_{\mathbf{n}} \left[h_x F_x + h_y F_y + h_z F_z \right],
\end{aligned} \tag{62}$$

with (a, b, c, d) from Eq. (98) and the Bertaut combinations Λ_X^a defined in Eq. (25). Three features of Eq. (62) are worth pointing out:

- Every Tm coupling sits on the Bertaut combination whose σ -vector matches the field's global axis (Klein-four selection rule). *No uniform Tm-sector coupling carries σ_F* , so the homogeneous Λ_F^a combinations cannot be excited by the bare linear drive.
- Fe couples to the uniform F_a combination only (per Eq. (35) the global Fe Zeeman acts as $-g_e\mu_B \sum_a h_a F_a$ once the local-frame sum is performed, because each $h_a S_\mu^a = h_a \eta_\mu^a \tilde{S}_\mu^a$ projects onto the σ_F -symmetric piece of $\tilde{S}_\mu^a$). At $\mathbf{q} = 0$ this is exactly the F_x, F_y, F_z ferromagnetic dipole.
- The ratio of Tm-electric to Tm-magnetic coupling at fixed wave impedance $|\mathbf{E}|/|\mathbf{H}| = Z_0$ is $|p|/(g_J\mu_B Z_0 |\mathbf{J}^{\text{eff}}|)$, controlling which channel dominates each pathway.

The two propagation geometries Eq. (61) now read off directly:

- $H \parallel \hat{a}, E \parallel \hat{c}$: the drive populates the four *distinct* Bertaut combinations Λ_C^5, Λ_C^7 (axial h_x) and Λ_G^4, Λ_G^6 (polar e_z). All four sit in $\Gamma_3 + \Gamma_4$ (off the Γ_2 ground state), hence drive the ω_{13}, ω_{23} Tm coherences.
- $H \parallel \hat{c}, E \parallel \hat{a}$: the drive populates Λ_G^2 (axial h_z , the unique Γ_2 admixture — this is the qAFM channel) and Λ_C^1 (polar e_x , in Γ_4). The Λ_G^2 coupling is the Tm partner of the Fe G_z Néel coordinate, which is why this geometry mixes Tm and Fe through $\tilde{\chi}^{2z}$.

VII. MICROSCOPIC HAMILTONIAN AND $\mathbf{q} = 0$ REDUCTION

At this point the microscopic Hamiltonian derivation is complete. The bond-resolved time-dependent Hamiltonian is

$$H_{\text{micro}}(t) = H_{\text{Fe}} + H_{\text{CEF}}^{\text{Tm, 4-sl}} + H_{\text{Tm-Fe}}^{\text{lin}} + H_{\text{drive}}^{\text{m}}(t) + H_{\text{drive}}^{\text{e}}(t) \tag{63}$$

with the Fe four-sublattice Hamiltonian H_{Fe} from Eq. (30), the four-sublattice Tm crystal-field Hamiltonian $H_{\text{CEF}}^{\text{Tm, 4-sl}}$ from Eq. (14), the time-reversal-even linear Fe-Tm coupling $H_{\text{Tm-Fe}}^{\text{lin}}$ from Eq. (50), and the THZ

magnetic/electric drive terms $H_{\text{drive}}^{\text{m,e}}(t)$ from Eqs. (59) and (60).

For the uniform low-temperature spectroscopy problem studied here, we then integrate out the virtual Tm processes at second order and keep the resulting $\mathbf{q} = 0$ Van Vleck back-action functional $H_{\text{aniso-Tm}}$. The working uniform-sector Hamiltonian used in the rest of the manuscript is therefore

$$H_{\text{total}}(t) = H_{\text{micro}}(t) + H_{\text{aniso-Tm}}, \tag{64}$$

where $H_{\text{aniso-Tm}}$ is the effective $\mathbf{q} = 0$ second-order back-action functional of Eq. (55). Equation (63) is the microscopic all-bond Hamiltonian; Eq. (64) is the derived working Hamiltonian for the uniform spectroscopy problem.

For the uniform $\mathbf{q} = 0$ response analyzed below, the drive sector is further reduced to the Bertaut form Eq. (62), and the static Tm part is projected to the three-level qutrit Hamiltonian $H_{\text{Tm}}^{(0)}$ of Eq. (52). The subsequent low-temperature Γ_2 analysis then linearizes Eq. (64) around the background $G_z + F_x$, keeps the low-frequency Fe block $M_{4 \times 4}^{\Gamma_2}$, and combines it with the Tm SU(3) block and the surviving $\mathbf{q} = 0$ coupling channels. Dissipation is added only at the equation-of-motion level through the Lindblad superoperator and is not part of $H_{\text{total}}(t)$.

VIII. 2DCS FRAMEWORK AND EXPERIMENTAL GEOMETRIES

A. Experimental nonlinear field, normal coordinates, and geometry channels

Experimentally the nonlinear field is isolated by subtracting the single-pulse controls from the full two-pulse response,

$$E_{\text{sig}}^{\text{NL}}(t_3, t_2, t_1) = E_{AB}(t_3, t_2, t_1) - E_A(t_3, t_2, t_1) - E_B(t_3, t_2, t_1). \tag{65}$$

Equation (65) is the directly measured time-domain quantity; the response-function formalism below is its perturbative evaluation in terms of the Hamiltonian Eq. (64) and the emission operator Eq. (83).

For a standard magnetic-2DCS presentation it is convenient to use the collective normal coordinates of the

low-temperature Γ_2 phase,

$$\begin{aligned} Q_{\text{qFM}}(t) &\equiv q_{\text{qFM}}(t), & \Omega_{\text{qFM}} &= \omega_{\text{qFM}}, \\ Q_{\text{qAFM}}(t) &\equiv q_{\text{qAFM}}(t), & \Omega_{\text{qAFM}} &= \omega_{\text{qAFM}}, \\ Q_{12}(t) &\equiv 2 \Re \rho_{12}(t), & \Omega_{12} &= \omega_{12}, \\ Q_{13}(t) &\equiv 2 \Re \rho_{13}(t), & \Omega_{13} &= \omega_{13}, \\ Q_{23}(t) &\equiv 2 \Re \rho_{23}(t), & \Omega_{23} &= \omega_{23}, \end{aligned} \quad (66)$$

where Q_{13} and Q_{23} mean the Bertaut-projected 13 and 23 coherences selected by the active geometry, i.e. the appropriate member of the set written explicitly in Eq. (111). At linear order these coordinates obey the standard coupled-oscillator form

$$\ddot{Q}_\mu + 2\Gamma_\mu \dot{Q}_\mu + \Omega_\mu^2 Q_\mu + \sum_\nu \kappa_{\mu\nu} Q_\nu = F_\mu(t), \quad (67)$$

with the retained Fe-Tm mixing contained in $\kappa_{12,\text{qAFM}} \sim \kappa_{\text{qAFM},12} \sim \chi^{2z} \sin \phi_{\text{qAFM}}$ and the remaining off-diagonal entries fixed by the symmetry analysis of Secs. VI A and XII.

The geometry-resolved drive and detection channels are then:

- Geometry I ($H \parallel a, E \parallel c$, VV): the pump drives the mirror-odd Tm sector $\Lambda_C^5, \Lambda_C^7, \Lambda_G^4, \Lambda_G^6$; VV detection projects onto the same mirror-odd channel, so the directly bright linear coordinates are Q_{13} and Q_{23} .
- Geometry II ($H \parallel a, E \parallel c$, VH): the pump is identical to Geometry I, but VH detection projects onto the mirror-even channel Λ_C^2, Λ_C^1 ; therefore Q_{12} is read out only after Fe-to-Tm conversion through the qAFM admixture.
- Geometry III ($H \parallel c, E \parallel a$, VH): the pump drives Λ_C^2, Λ_C^1 , and the Fe F_z Zeeman channel; the directly addressed linear coordinates are Q_{12} and Q_{qFM} , while VH detection reads the orthogonal qAFM emission channel.

Each observed peak must therefore be explained twice: once as an allowed Liouville pathway and once as a symmetry-allowed coupling among the normal coordinates Q_μ .

B. The third-order response function

The 2DCS signal in the rephasing/non-rephasing direction is

$$\begin{aligned} S(\omega_t, t_2, \omega_\tau) &= \int_0^\infty dt_3 e^{i\omega_t t_3} \int_0^\infty dt_1 e^{\pm i\omega_\tau t_1} \\ &\quad \times R^{(3)}(t_3, t_2, t_1), \end{aligned} \quad (68)$$

where the $\pm$ selects rephasing ($-$) or non-rephasing ($+$) and

$$\begin{aligned} R^{(3)}(t_3, t_2, t_1) &= \left(\frac{i}{\hbar}\right)^3 \text{Tr} \left\{ \hat{O}_{\text{det}} \mathcal{G}(t_3) \mathcal{V}_3 \mathcal{G}(t_2) \right. \\ &\quad \left. \times \mathcal{V}_2 \mathcal{G}(t_1) \mathcal{V}_1 \rho_{\text{eq}} \right\}, \end{aligned} \quad (69)$$

with $\mathcal{V}_i = [\hat{V}_i, \cdot]$ the dipole-coupling superoperator at interaction i , $\mathcal{G}(t) = e^{-i\mathcal{L}t}$ the free-evolution superoperator (with damping), and $\hat{O}_{\text{det}}$ the dipole detected in emission. Each Liouville pathway is one of the eight double-sided Feynman diagrams that arise from the choice of left/right action of $\mathcal{V}_i$ on the density matrix.

IX. SU(3) BLOCH DYNAMICS WITH CPTP DISSIPATION

A. Coherent evolution

The Tm density matrix on each sublattice obeys

$$\dot{\rho}_{\mathbf{n}}^{(j)} = -i[H_{\mathbf{n}}^{(j)}, \rho_{\mathbf{n}}^{(j)}] + \mathcal{L}^{\text{diss}}[\rho_{\mathbf{n}}^{(j)}], \quad (70)$$

where $H_{\mathbf{n}}^{(j)}$ is the local Tm projection of the complete Hamiltonian Eq. (64) (bare CEF + linear Fe coupling + drive + Van Vleck back-action). In the SU(3) Bloch representation $\rho = \mathbb{1}/3 + \frac{1}{2} \sum_a n^a \lambda^a$ with $n^a = \langle \lambda^a \rangle$, the coherent equation reads

$$\dot{n}^a = \sum_{b,c} f^{abc} h^b n^c, \quad H = h_0 \mathbb{1} + \frac{1}{2} \sum_a h^a \lambda^a. \quad (71)$$

With this convention, the bare CEF field is $h^3 = 2h_3^{(0)}$, $h^8 = 2h_8^{(0)}$ (factor of 2 from the $\frac{1}{2}$ in the convention), giving the correct precession frequencies $\omega_{12} = 2|h_3^{(0)}|/\hbar = |\epsilon_2 - \epsilon_1|/\hbar$, etc. The factor-of-2 chain is collected in App. A and propagated consistently.

B. Lindblad dissipator

A naive pure-dephasing dissipator $\dot{n}^a \supset -\Gamma_a n^a$ is not in general CPTP-compliant on a qutrit because it can drive ρ outside the positive cone. We therefore use a secular Lindblad form in the CF eigenbasis with explicit relaxation jumps and diagonal dephasing projectors:

$$\begin{aligned} \mathcal{L}^{\text{diss}}[\rho] &= \sum_{\alpha > \beta} \Gamma_{\alpha \rightarrow \beta} \left[L_{\beta\alpha} \rho L_{\beta\alpha}^\dagger \right. \\ &\quad \left. - \frac{1}{2} \{ L_{\beta\alpha}^\dagger L_{\beta\alpha}, \rho \} \right] \\ &\quad + \sum_{\alpha=1}^3 \gamma_\alpha^\phi \left[P_\alpha \rho P_\alpha - \frac{1}{2} \{ P_\alpha, \rho \} \right], \end{aligned} \quad (72)$$

with $L_{\beta\alpha} = |E_\beta\rangle\langle E_\alpha|$ for inelastic relaxation and $P_\alpha = |E_\alpha\rangle\langle E_\alpha|$ for pure dephasing. The relaxation rates obey detailed balance $\Gamma_{\alpha \rightarrow \beta}/\Gamma_{\beta \rightarrow \alpha} = e^{-(\epsilon_\alpha - \epsilon_\beta)/k_B T}$, while the coherence $|E_\alpha\rangle\langle E_\beta|$ acquires the additional dephasing rate $(\gamma_\alpha^\phi + \gamma_\beta^\phi)/2$. CPTP is automatic for any $\Gamma_{\alpha \rightarrow \beta}, \gamma_\alpha^\phi \geq 0$.

C. Linearization around the $|E_1\rangle$ vacuum

At $T \ll (\epsilon_2 - \epsilon_1)/k_B \approx 22$ K the Tm population is concentrated in $|E_1\rangle$, $\rho \approx |E_1\rangle\langle E_1|$, and the equilibrium SU(3) Bloch vector is

$$n^{3,\text{eq}} = +1, \quad n^{8,\text{eq}} = +1/\sqrt{3}, \quad n^{a,\text{eq}} = 0 \text{ otherwise.} \quad (73)$$

Linearizing Eq. (71) around Eq. (73) with the bare CEF field of Eq. (52) block-diagonalizes into three 2×2 oscillator blocks:

$$\begin{aligned} (\delta n^1, \delta n^2) : \quad \omega_{12} &= |\epsilon_2 - \epsilon_1|/\hbar, \\ (\delta n^4, \delta n^5) : \quad \omega_{13} &= |\epsilon_3 - \epsilon_1|/\hbar, \\ (\delta n^6, \delta n^7) : \quad \omega_{23} &= |\epsilon_3 - \epsilon_2|/\hbar, \end{aligned} \quad (74)$$

plus the two zero-frequency population channels $(\delta n^3, \delta n^8)$.

Validity regime. Eq. (73) is well-justified for $T \lesssim 20$ K. At the spin-reorientation temperatures $T_{\text{SR}1,2} \approx 81\text{--}91$ K the populations are $p_1 \approx 0.50$, $p_2 \approx 0.40$, $p_3 \approx 0.10$ from Boltzmann weighting, and the linearization around $|E_1\rangle$ is *not valid*; the model must be re-linearized around a thermally mixed reference state, with the equilibrium n^a shifted accordingly. We restrict the present analysis to $T \lesssim 20$ K where the standard linearization holds.

X. EFFECTIVE FE MAGNON SECTOR AROUND THE Γ_2 GROUND STATE

A. Linearization

We write

$$\begin{aligned} \mathbf{F} &= F_x \hat{x} + \delta \mathbf{F}, \quad \mathbf{G} = G_z \hat{z} + \delta \mathbf{G}, \\ \mathbf{C} &= C_y \hat{y} + \delta \mathbf{C}, \quad \mathbf{A} = \delta \mathbf{A}, \end{aligned} \quad (75)$$

with the spin-length constraint $|\mathbf{S}_\mu| = S$ enforced through $\mathbf{F} \cdot \delta \mathbf{F} + \mathbf{G} \cdot \delta \mathbf{G} + \mathbf{C} \cdot \delta \mathbf{C} = 0$ order by order. To leading order this kills three of the twelve components $(\delta \mathbf{F}, \delta \mathbf{G}, \delta \mathbf{C}, \delta \mathbf{A})$, leaving nine independent linearized degrees of freedom. Of these, four split into the high-frequency optical branches (involving $\delta \mathbf{A}$, δC_z , etc.) at the exchange scale, and the remaining *five* couple via the DM and anisotropy terms. With the corrected DM Bertaut Hamiltonian Eq. (46), the canonical Treves–Bertaut Lifshitz invariant $D_1 F_x G_z$ is carried inside the reduced $(\mathbf{F}, \mathbf{G}, \mathbf{C}, \mathbf{A})$ manifold, so the low-frequency block closes on the four-component subspace below to leading order in $D_{1,2}/J_1$; the exchange-stiff Γ_4 partners $(\delta A_x, \delta A_z, \delta C_x, \delta C_z)$ only renormalize the κ coefficients of Eq. (77). For spectroscopy we introduce the *effective* low-frequency coordinate vector

$$\mathbf{v} \equiv (\delta F_y, \delta G_y; \delta F_z, \delta G_x)^T \quad (76)$$

whose last two entries represent the dominant projection of the qFM-like resonance (the soft *ac*-plane rotation),

while the first two represent the dominant projection of the qAFM-like hard-axis fluctuation. Any additional components of the exact eigenvectors are absorbed into the renormalized coefficients of the reduced model or into small phenomenological mixing angles introduced later.

B. Explicit 4×4 linearized matrix

In the effective basis Eq. (76), we therefore use an effective reduced Fe magnon block,

$$\dot{\mathbf{v}} = M_{4 \times 4}^{\Gamma_2} \mathbf{v}, \quad M_{4 \times 4}^{\Gamma_2} \equiv \begin{pmatrix} 0 & \kappa_{\text{qAFM}}^{(1)} & 0 & 0 \\ -\kappa_{\text{qAFM}}^{(2)} & 0 & 0 & 0 \\ 0 & 0 & 0 & \kappa_{\text{qFM}}^{(1)} \\ 0 & 0 & -\kappa_{\text{qFM}}^{(2)} & 0 \end{pmatrix}, \quad (77)$$

whose coefficients are the renormalized frequency scales obtained after eliminating the exchange-stiff partner modes omitted from Eq. (76). The previously printed α_i, β_i expressions are withdrawn: they are not the linearization of the corrected Section-IV Hamiltonian. In practice the κ 's are determined either from a numerical linearization of the full Fe sector or by fitting the two observed Fe resonance frequencies. The two 2×2 blocks encode, by construction, the dominant qAFM and qFM resonances used in the later reduced joint model.

C. qFM frequency

The qFM-like block $(\delta F_z, \delta G_x)$ represents the soft *ac*-plane rotation of the Γ_2 state. Its reduced-model frequency is

$$\omega_{\text{qFM}}^2 = \kappa_{\text{qFM}}^{(1)} \kappa_{\text{qFM}}^{(2)}. \quad (78)$$

After the exchange-stiff partner modes are eliminated, the familiar orthoferrite scaling is recovered only at the level of a schematic two-mode reduction,

$$\omega_{\text{qFM}}^2 \propto H_E^{\text{exch}} H_A^{\text{qFM}}, \quad (79)$$

where H_E^{exch} is the exchange scale and H_A^{qFM} the effective anisotropy scale controlling the soft *ac*-plane rotation, with $H_A^{\text{qFM}} \rightarrow 0$ at $T_{\text{SR}1}$. Thus qFM is the soft mode of the $\Gamma_2 \rightarrow \Gamma_{24}$ spin reorientation. In what follows we use ω_{qFM} as the experimentally calibrated or numerically linearized low-energy Fe resonance, not as the value of the withdrawn raw 4×4 formulas.

D. qAFM frequency: with the DM gap

The qAFM-like block $(\delta F_y, \delta G_y)$ represents the harder *b*-axis Fe resonance in the reduced model, with

$$\omega_{\text{qAFM}}^2 = \kappa_{\text{qAFM}}^{(1)} \kappa_{\text{qAFM}}^{(2)}. \quad (80)$$

After the same elimination of stiff partner modes, its gap is governed schematically by hard-axis anisotropy plus a DM-induced contribution,

$$\omega_{\text{qAFM}}^2 \sim H_E^{\text{exch}} \left(H_A^{\text{qAFM}} + c_D \frac{H_D^2}{H_E^{\text{exch}}} \right), \quad (81)$$

where H_A^{qAFM} is the effective hard-axis anisotropy scale, H_D the effective DM scale, and $c_D = O(1)$ depends on the precise reduction convention. The important qualitative point is that the qAFM resonance remains finite across T_{SR1} because the DM sector produces a nonzero effective gap even when the bare hard-axis anisotropy is small. Later sections encode the small projection of this mode onto the particular G -staggered coordinate that couples to the Tm λ^2 channel by a phenomenological mixing angle ϕ_{qAFM} , rather than reading it directly off the withdrawn raw block formulas.

E. Spin-reorientation transitions

The signs of the anisotropy differences $K_c - K_a$, $K_c - K_b$ depend on temperature through the rare-earth contribution $K_{ab}^{\text{Tm}}(T)$ to the effective Fe single-ion anisotropy (Sec. VD). In TmFeO₃, $K_c - K_a$ changes sign at $T_{\text{SR1}} \approx 81$ K and $K_c - K_b$ changes sign at $T_{\text{SR2}} \approx 91$ K, giving:

- $T < T_{\text{SR1}}$: Γ_2 phase, $\mathbf{G} \parallel \hat{z}$, soft mode is qFM ($\omega_{\text{qFM}} \rightarrow 0$ as $T \rightarrow T_{\text{SR1}}$);
- $T_{\text{SR1}} < T < T_{\text{SR2}}$: Γ_{24} mixed phase, with $\mathbf{G}$ continuously rotating in the ac plane; both the qFM-like and qAFM-like modes have intermediate frequencies;
- $T > T_{\text{SR2}}$: Γ_4 phase, $\mathbf{G} \parallel \hat{x}$, soft mode is qFM-like with respect to Γ_4 ($\omega \rightarrow 0$ as $T \rightarrow T_{\text{SR2}}^+$).

This is the canonical orthoferrite spin-reorientation phenomenology with two second-order transitions bracketing one continuous rotation, distinct from a single SR transition picture [9–11].

XI. COHERENT EMISSION

In SI units, the radiated electric field at the detector contains both the electric-dipole polarization $\mathbf{P}$ and the magnetic-dipole magnetization $\mathbf{M}$, related by the impedance of free space:

$$\mathbf{E}_{\text{sig}}(t, \mathbf{r}) \propto \partial_t \mathbf{P}_\perp(t, \mathbf{r}) + \frac{Z_0}{c} \partial_t \mathbf{M}_\perp(t, \mathbf{r}) \times \hat{\mathbf{k}}, \quad (82)$$

where $Z_0 = \sqrt{\mu_0/\epsilon_0} \approx 377 \Omega$ is the free-space impedance, and the perpendicular projection is with respect to the propagation direction $\hat{\mathbf{k}}$. In SI the prefactor on the $\mathbf{M}$ contribution is $Z_0/c = \mu_0$, but writing it as Z_0/c emphasizes that the relative weight of electric- and

magnetic-dipole emission in the detected field is set by the impedance.

The signal is the projection of Eq. (82) onto the detector polarization $\hat{\mathbf{e}}_{\text{det}}$, summed coherently over the illuminated volume. For a slab of thickness L (with $L \ll \lambda_{\text{THz}}$ in the thin-sample limit) and uniform pump,

$$E_{\text{sig}}(t) \propto L \hat{\mathbf{e}}_{\text{det}} \cdot \left[\partial_t \mathbf{P}(t) + \frac{Z_0}{c} \partial_t \mathbf{M}(t) \times \hat{\mathbf{k}} \right], \quad (83)$$

where the cell-averaged polarization and magnetization are

$$\begin{aligned} \mathbf{P} &= \frac{1}{V_c} \sum_j \langle \mathbf{d}_{\mathbf{n},j}^{\text{eff}} \rangle, \\ \mathbf{M} &= \frac{1}{V_c} \left[g_e \mu_B \sum_i \langle \mathbf{S}_{\mathbf{n},i} \rangle \right. \\ &\quad \left. + g_J \mu_B \sum_j \langle \mathbf{J}_{\mathbf{n},j}^{\text{eff}} \rangle \right], \end{aligned} \quad (84)$$

and $V_c = abc$ is the unit-cell volume. In the local-frame formulation, $\langle \mathbf{J}_{\mathbf{n},j}^{\text{eff}} \rangle$ and $\langle \mathbf{d}_{\mathbf{n},j}^{\text{eff}} \rangle$ must be rotated back to the global frame using R_μ^{axial} and R_μ^{polar} respectively before summation (Sec. IIC).

XII. LINEARIZED JOINT EOM

A. Joint variable vector

The joint dynamics combines the Fe magnon block (4 dofs in the low-frequency subspace, Eq. (76)) with the Tm SU(3) Bloch block (6 dofs in the three coherence pairs, Eq. (74)):

$$\mathbf{w} = (\delta F_y, \delta G_y, \delta F_z, \delta G_x; \delta n^1, \delta n^2, \delta n^4, \delta n^5, \delta n^6, \delta n^7)^T. \quad (85)$$

The two Tm population channels ($\delta n^3, \delta n^8$) decouple at zeroth order and enter only via second-order Van Vleck back-action (Sec. VD).

B. Joint matrix

The linearized EOM is $\dot{\mathbf{w}} = M^{\Gamma_2, \text{joint}} \mathbf{w}$ with $M^{\Gamma_2, \text{joint}}$ block-structured as

$$M^{\Gamma_2, \text{joint}} = \begin{pmatrix} M_{4 \times 4}^{\Gamma_2} & M_{4 \times 6}^{\text{Fe} \rightarrow \text{Tm}} \\ M_{6 \times 4}^{\text{Tm} \rightarrow \text{Fe}} & M_{6 \times 6}^{\text{Tm}} \end{pmatrix}, \quad (86)$$

with:

- $M_{4 \times 4}^{\Gamma_2}$: the Fe magnon matrix of Eq. (77);
- $M_{6 \times 6}^{\text{Tm}}$: block-diagonal in the three CF doublets, with $\omega_{12,13,23}$ on the diagonals;

- $M_{4 \times 6}^{\text{Fe} \rightarrow \text{Tm}}$: after projecting the omitted δG_z coordinate onto the low-frequency qAFM eigenvector, the only surviving reduced-basis entry is the effective qAFM $\rightarrow \delta n^2$ coupling with strength $\propto \bar{\chi}^{2z} \sin \phi_{\text{qAFM}}$;
- $M_{6 \times 4}^{\text{Tm} \rightarrow \text{Fe}}$: the reciprocal $\delta n^2 \rightarrow$ qAFM entry generated by the same projected exchange vertex.

Thus the reduced 10-component model contains no bare δG_z row or column. The Fe–Tm hybridization enters only through the projected coupling

$$\delta n^2(t) \leftrightarrow \bar{\chi}^{2z} \sin \phi_{\text{qAFM}} \text{qAFM}(t) \quad (87)$$

with $\sin \phi_{\text{qAFM}} = \mathcal{O}(D_1/J_1) \ll 1$. This is the parametric origin of the (qAFM, E_{12}) peak in Geometry II.

XIII. DETAILED LIOUVILLE-PATHWAY ANALYSIS OF THE SIX 2DCS PEAKS

A. General peak amplitude expression

For a peak generated by a Liouville pathway $\mathcal{P}$ that carries a coherence during t_1 , an intermediate coherence or population $\eta_{\mathcal{P}}$ during t_2 , and a radiating coherence during t_3 , the pathway contribution has the generic form

$$A_{\mathcal{P}}(\omega_{\tau}, t_2, \omega_t) = \left(\frac{i}{\hbar}\right)^3 \rho_{00} \frac{\langle d | \hat{O}_{\text{det}} | c \rangle \mathcal{V}_{\mathcal{P}}^{(3)} \mathcal{V}_{\mathcal{P}}^{(2)} \mathcal{V}_{\mathcal{P}}^{(1)}}{\mathcal{D}_{\tau, \mathcal{P}}(\pm \omega_{\tau}) \mathcal{D}_{t, \mathcal{P}}(\omega_t)} e^{-[\Gamma_{\eta, \mathcal{P}} + i\Omega_{\eta, \mathcal{P}}]t_2}, \quad (88)$$

where $\mathcal{V}_{\mathcal{P}}^{(i)}$ are the pathway-dependent matrix elements or effective bilinear vertices selected by the three interactions, $\mathcal{D}_{\tau, \mathcal{P}}$ and $\mathcal{D}_{t, \mathcal{P}}$ are the first- and third-interval coherence denominators, and $\Omega_{\eta, \mathcal{P}} = 0$ when the intermediate object is a population grating. The explicit peak amplitudes written below are specializations of Eq. (88) to the six symmetry-allowed pathways. There is also a formal population branch at $(\omega_{\tau}, \omega_t) = (\omega_{12}, 0)$: the first interaction creates ρ_{12} , the second converts it into the diagonal SU(3) population channels $\Lambda_F^{3,8} \sim (n_1, n_2)$ through Eqs. (121)–(122), and the third interval then evolves at zero frequency. We keep this $(E_{12}, 0)$ pathway in mind as a real population feature of the Liouville problem, but do not include it in the radiative peak count because $\hat{O}_{\text{det}}$ is a dipole operator and the detected field is proportional to $\partial_t \mathbf{P}$ and $\partial_t \mathbf{M}$ [Eqs. (82)–(83)]; a static population channel therefore carries no coherent far-field THz emission at $\omega_t = 0$ unless one extends the measurement model to direct population readout or relaxation-tail detection. Rephasing pathways use the minus sign in $\mathcal{D}_{\tau, \mathcal{P}}(\pm \omega_{\tau})$ and produce diagonal-axis peaks; non-rephasing pathways use the plus sign and produce cross-axis peaks. We work out the six peaks of the corrected list explicitly.

B. Bertaut bookkeeping for every pathway

Before working out the six peaks individually, we record the σ -vector that each vertex carries, and the

global Klein-four selection rule that constrains every double-sided Feynman diagram. Each interaction vertex $\hat{V}_i$ in Eq. (88) carries a σ -vector inherited from the field component and the Bertaut combination it excites (Eqs. (62)); a coherent pathway is allowed only if the product of σ -vectors along the diagram returns to σ_F (the trivial irrep), since the detected one-cell observable transforms as σ_F .

a. σ -rule for a third-order pathway. Writing σ_{V_i} for the σ -vector of vertex $\hat{V}_i$ and $\sigma_{O_{\text{det}}}$ for the detector,

$$\sigma_{V_1} \cdot \sigma_{V_2} \cdot \sigma_{V_3} \cdot \sigma_{O_{\text{det}}} \stackrel{!}{=} \sigma_F, \quad (89)$$

where the product is component-wise (Klein-four multiplication, Eq. (38)). This is the four-sublattice generalization of the conventional $\Gamma_{\text{det}} \subset \Gamma_{V_1} \otimes \Gamma_{V_2} \otimes \Gamma_{V_3}$ group-theoretic selection rule; for $Pbnm$ orthoferrites at $\mathbf{q} = 0$ it reduces to the requirement that the four-cycle of σ -vectors multiply to $(+, +, +, +)$.

b. σ -vectors of the six peaks. Reading off Eq. (62) and the matrix-element catalogue Eq. (98) (each $\mathbf{J}^{\text{eff}}$, $\mathbf{d}^{\text{eff}}$ vertex inherits the σ of its field index; each Fe Bertaut excitation F_a, G_a , etc., carries its own σ):

Peak	σ_{V_1}	σ_{V_2}	σ_{V_3}	$\sigma_{O_{\text{det}}}$	$\prod \sigma$	Bertaut path
(E_{12}, E_{23})	σ_C	σ_G	σ_C	σ_G	σ_F	$\Lambda_C^5 \Lambda_G^2 \Lambda_C^7 / \Lambda_G^2$
(E_{12}, E_{13})	σ_C	σ_G	σ_G	σ_C	σ_F	$\Lambda_C^5 \Lambda_G^2 \Lambda_G^2 / \Lambda_C^5$
(E_{12}, qAFM)	σ_G	σ_G	σ_G	σ_G	σ_F	$\Lambda_G^2 \Lambda_G^2 G_z$
(qAFM, E_{12})	σ_F	σ_F	σ_G	σ_G	σ_F	$F_x F_x (\sin \phi_{\text{qAFM}} G_z) \Lambda_G^2$
(qAFM, E_{13})	σ_F	σ_F	σ_C	σ_C	σ_F	$F_x F_x (\sin \phi_{\text{qAFM}} G_z) \Lambda_C^5$
$(\text{qFM}, \text{qAFM})$	σ_F	σ_F	σ_F	σ_G	$\sigma_G \rightarrow \sigma_F$ via $T^{(2)}$	$F_y F_y (T^{(2)}: \sigma_F \rightarrow \sigma_G) G_z$

(90)

For the first five rows the four-cycle of σ -vectors collapses directly to σ_F via the Klein algebra $\sigma_C \sigma_C = \sigma_G \sigma_G = \sigma_F$, etc. For the last row (qFM, qAFM) the $T^{(2)}$ vertex carries an explicit $\sigma_F \rightarrow \sigma_G$ transformation (DM-induced $F \leftrightarrow G$ admixture, App. E), supplying the final σ_G needed to balance the detector.

c. Consequences for the pathway analysis. The selection-rule Eq. (89) explains, without explicit computation, why every cross-sector peak that links Fe to Tm requires either (a) the small DM-induced $\sin \phi_{\text{qAFM}}$ admixture of G_z into the qAFM eigenvector ((qAFM, $E_{12,13}$) and (E_{12} , qAFM)), or (b) the $T^{(2)}$ DM-driven $F \leftrightarrow G$ Bertaut mixing (qFM, qAFM). No “ σ -orphan” peak — one whose σ -vector product would fail to close to σ_F — appears in the six-peak list, and the selection rule predicts that such peaks have strictly zero amplitude at $\mathbf{q} = 0$.

C. Geometry I: $H \parallel a, E \parallel c, \mathbf{V}\mathbf{V}$

The pump and detect interactions are $\hat{V} = h_x \mathbf{J}_x^{\text{eff}} + e_z \mathbf{d}_z^{\text{eff}}$, which couples to $\lambda^{4,5,6,7}$ (mirror-odd Tm sector, $A_{1,2}^-$). VV detection reads out the same combination.

a. (E_{12}, E_{23}). Two pump interactions $V_1 = V_2 \in \{\lambda^4, \lambda^5, \lambda^6, \lambda^7\}$ generate the $|E_2\rangle\langle E_1|$ sector only at *second order*: the odd-sector bilinears (4, 6) and (5, 7) close onto λ^1 by anticommutation, while (4, 7) and (5, 6) close onto λ^2 by the complementary commutator/anticommutator channel. We denote the resulting effective quadratic Geometry-I source on the E_{12} coherence by $\mathcal{S}_{12}^{\text{odd}}$. The third odd-sector interaction then promotes $|E_2\rangle$ to $|E_3\rangle$, producing emission at ω_{23} . Amplitude:

$$A_{(E_{12}, E_{23})} \propto \frac{\mathcal{S}_{12}^{\text{odd}} \cdot M_{13} \cdot M_{23}}{(\Gamma_{12} + i\omega_{12})(\Gamma_{23} + i\omega_{23})} e^{-\Gamma_{23} t_2} \rho_{00}^{\text{Tm}}, \quad (91)$$

where $M_{\alpha\beta} \equiv \langle E_\alpha | \mathbf{J}_x^{\text{eff}} | E_\beta \rangle$ and $\mathcal{S}_{12}^{\text{odd}}$ is the effective second-order source built from the allowed Geometry-I odd-sector vertices $\{h_x J_x^{\text{eff}}, e_z d_z^{\text{eff}}\}$.

b. (E_{12}, E_{13}). Same E_{12} excitation as above; the third interaction $V_3 \sim \lambda^5$ promotes $|E_2\rangle$ to $|E_3\rangle$ via the alternative odd-sector matrix element. In the magnetic- x readout channel the amplitude is analogous to Eq. (91)

with $M_{23} \rightarrow M_{13}$:

$$A_{(E_{12}, E_{13})}/A_{(E_{12}, E_{23})} = M_{13}/M_{23} = 2.392/0.913 \approx 2.6. \quad (92)$$

This ratio is a parameter-free prediction of the model.

c. (qAFM, E_{13}). Fe-pumped pathway: the magnetic field h_x excites the qAFM mode via the Fe Zeeman vertex; the qAFM eigenvector has a $\sin \phi_{\text{qAFM}} \sim D_1/J_1$ admixture along δG_z (Sec. XD); the δG_z coordinate sources the linear $\bar{\chi}^{2z} \delta G_z \lambda^2$ exchange field on the Tm qutrit, populating $|E_2\rangle\langle E_1|$ through the SU(3) generator λ^2 . A subsequent odd-sector interaction promotes to $|E_3\rangle$, producing E_{13} emission. Amplitude is suppressed by the small ratio D_1/J_1 relative to the purely Tm pathways:

$$A_{(\text{qAFM}, E_{13})} \propto (D_1/J_1) \cdot \bar{\chi}^{2z} \cdots, \quad (93)$$

making it the weakest of the four Geometry-I peaks.

D. Geometry II: $H \parallel a, E \parallel c, \mathbf{V}\mathbf{H}$

The pump is identical to Geometry I (mirror-odd Tm sector); VH detection reads out the orthogonal mirror-even sector via $\lambda^1 \sim d_x^{\text{eff}}$ and $\lambda^2 \sim J_z^{\text{eff}}$.

a. (qAFM, E_{12}). The cleanest linear-exchange pathway in the dataset: $h_x \rightarrow \text{qAFM} \rightarrow (D_1/J_1) \delta G_z \rightarrow \bar{\chi}^{2z} \delta n^2 \rightarrow E_{12}$ emission via $J_z^{\text{eff}} \sim \lambda^2$. Amplitude:

$$A_{(\text{qAFM}, E_{12})} \propto (g_e \mu_B)^2 (D_1/J_1) \bar{\chi}^{2z} |M_{12}| \frac{1}{(\Gamma_{\text{qAFM}} + i\omega_{\text{qAFM}})(\Gamma_{12} + i\omega_{12})}. \quad (94)$$

This is a third-order signal (two Fe Zeeman interactions and one Tm emission) with three causal time orderings. The pure-Fe-pump character makes it especially clean to identify experimentally.

E. Geometry III: $H \parallel c, E \parallel a, \mathbf{V}\mathbf{H}$

The pump directly addresses the even Tm sector via $h_z J_z^{\text{eff}} \sim h_z \lambda^2$ and $e_x d_x^{\text{eff}} \sim e_x \lambda^1$.

a. (E_{12} , qAFM). Reciprocal of Geometry-II pathway: the pump creates δn^2 directly; this drives δG_z via

$\bar{\chi}^{2z}$; δG_z couples into the qAFM eigenvector through its DM admixture; qAFM emits during t_3 . Amplitude:

$$A_{(E_{12}, \text{qAFM})} \propto (g_J \mu_B)^2 \cdot |M_{12}|^2 \cdot \bar{\chi}^{2z} \cdot (D_1/J_1) \cdot (g_e \mu_B), \quad (95)$$

which is the time-reversed pair of Eq. (94).

b. (qFM, qAFM). Fe nonlinearity: the magnetic field h_z couples directly to the qFM coordinate via the Zeeman vertex; the qFM and qAFM modes are coupled at second order in the LLG dynamics through the spin-length constraint and the DM-induced mixing of $\mathbf{F}$ and $\mathbf{G}$. The bilinear coupling tensor $T_{\text{qFM} \rightarrow \text{qAFM}}^{(2)}$ is the effective quadratic Fe vertex inherited from the second-order Bertaut expansion (App. E):

$$T_{\text{qFM} \rightarrow \text{qAFM}}^{(2)} \equiv \Omega_{\text{qFM}, \text{qAFM}}^{(2)} \quad (96)$$

whose symmetry is fixed by the DM-induced $F \leftrightarrow G$ mixing and whose magnitude scales linearly with the small canting ratio D_1/J_1 in a reduced two-sublattice description. The resulting peak amplitude is $\propto T^{(2)} \cdot (g_e \mu_B)^2$. Fe matrix elements. This peak is purely a Fe-sector nonlinearity and does not require Tm involvement.

F. Summary peak table

The reciprocal pair Geometry II (qAFM, E_{12}) and Geometry III (E_{12} , qAFM) measure the same exchange vertex $\bar{\chi}^{2z}$ in the two opposite time orderings; their amplitude ratio in the rephasing vs. non-rephasing channels is a stringent self-consistency test of the model.

G. Explicit matrix-element catalogue

We collect here, for every one of the six Liouville pathways, the *four* operator factors that enter Eq. (88) (three vertices V_1, V_2, V_3 plus one detection $\hat{O}_{\text{det}}$), the two coherence denominators and the t_2 decay, in the same notation. Define

$$g_{\bar{e}} \equiv g_e \mu_B, \quad g_J \equiv g_J \mu_B, \\ \mathcal{D}_{ab}(\omega) \equiv \Gamma_{ab} + i(\omega - \omega_{ab}), \quad \mathcal{D}_m(\omega) \equiv \Gamma_m + i(\omega - \omega_m), \quad (97)$$

for $m \in \{\text{qFM}, \text{qAFM}\}$. The numerical values of the projected Tm matrix elements (T-symmetric gauge, Eq. (20)) are

$$\begin{aligned} M_{12}^{(z)} &\equiv \langle E_1 | J_z^{\text{eff}} | E_2 \rangle = -5.264 i, \\ M_{13}^{(x)} &\equiv \langle E_1 | J_x^{\text{eff}} | E_3 \rangle = -2.392 i, \\ M_{23}^{(x)} &\equiv \langle E_2 | J_x^{\text{eff}} | E_3 \rangle = -0.913 i, \\ M_{13}^{(y)} &\equiv \langle E_1 | J_y^{\text{eff}} | E_3 \rangle = +2.787 i, \\ M_{23}^{(y)} &\equiv \langle E_2 | J_y^{\text{eff}} | E_3 \rangle = -0.466 i, \end{aligned} \quad (98)$$

all in units of $\hbar$. Diagonal-Cartan elements $\langle E_\alpha | J_a | E_\alpha \rangle$ vanish identically in the T-symmetric gauge for $\alpha = 1$ and are absorbed into $h_{3,8}^{(0)}$ for $\alpha = 2, 3$ (Eq. (53)). For the electric-dipole channel we likewise write $D_{12}^{(x)} \equiv \langle E_1 | d_x^{\text{eff}} | E_2 \rangle = p_x$, $D_{13}^{(z)} \equiv \langle E_1 | d_z^{\text{eff}} | E_3 \rangle = p_{z4}$, and $D_{23}^{(z)} \equiv \langle E_2 | d_z^{\text{eff}} | E_3 \rangle = p_{z6}$.

a. Pathway notation. For each peak we write the rephasing (R) and non-rephasing (NR) double-sided diagrams as

$$|0\rangle\langle 0| \xrightarrow{V_1} |a\rangle\langle 0| \xrightarrow{V_2} (|c\rangle\langle 0| \text{ or } |a\rangle\langle d|) \xrightarrow{V_3} |e\rangle\langle f| \xrightarrow{\hat{O}_{\text{det}}}, \quad (99)$$

where each arrow is a left- or right-multiplication by the indicated vertex. The Tm states are $\{|E_1\rangle \equiv |0\rangle, |E_2\rangle, |E_3\rangle\}$; the Fe magnon states are bosonic Fock states of the qFM and qAFM oscillators; the joint vacuum is $|0\rangle_{\text{Tm}} \otimes |\text{vac}\rangle_{\text{Fe}}$.

b. Geometry-I peaks. The pump-detect channels are $V_i \in \{h_x J_x^{\text{eff}}, e_z d_z^{\text{eff}}\} \propto \{\lambda^{5,7}, \lambda^{4,6}\}$ during t_1, t_3 . Because these odd-sector generators do not contain a direct λ^2 vertex, the E_{12} coherence in Geometry I is generated only at second order. We denote the resulting effective source by $\mathcal{S}_{12}^{\text{odd}}$; it collects the allowed odd-sector bilinears (4,6), (5,7), (4,7), and (5,6) that close onto $\lambda^{1,2}$. The four pathways:

TABLE III. Standard 2DCS pathway table for the six intrinsic peaks. Each row lists the excitation frequency ω_τ , the detection frequency ω_t , the symmetry or dynamical coupling that must be present, and the leading scaling of the corresponding peak amplitude.

Geometry	Peak	ω_τ	ω_t	Required coupling	Leading scaling
I (VV)	(E_{12}, E_{23})	ω_{12}	ω_{23}	Odd-sector Tm-only pathway with source $\mathcal{S}_{12}^{\text{odd}}$ and odd-sector 23 read-out	$\propto M_{13}M_{23} M_{12} ^2$
I (VV)	(E_{12}, E_{13})	ω_{12}	ω_{13}	Odd-sector Tm-only pathway with source $\mathcal{S}_{12}^{\text{odd}}$ and odd-sector 13 read-out	$\propto M_{13}^2 M_{12} ^2$
I (VV)	(qAFM, E_{13})	ω_{qAFM}	ω_{13}	DM admixture $\sin \phi_{\text{qAFM}}$ plus linear exchange $\bar{\chi}^{2z}$ converting qAFM motion into the even 12 channel before odd-sector promotion to E_{13}	$\propto (D_1/J_1)\bar{\chi}^{2z}M_{13}$
II (VH)	(qAFM, E_{12})	ω_{qAFM}	ω_{12}	Linear Fe $\rightarrow$ Tm conversion through $\bar{\chi}^{2z} \sin \phi_{\text{qAFM}}$ with VH readout in the mirror-even channel	$\propto (D_1/J_1)\bar{\chi}^{2z} M_{12} $
III (VH)	(E_{12}, qAFM)	ω_{12}	ω_{qAFM}	Reciprocal Tm $\rightarrow$ Fe conversion through the same $\bar{\chi}^{2z} \sin \phi_{\text{qAFM}}$ channel	$\propto M_{12} ^2\bar{\chi}^{2z}(D_1/J_1)$
III (VH)	$(\text{qFM}, \text{qAFM})$	ω_{qFM}	ω_{qAFM}	Fe-only quadratic mode-mixing vertex $T_{\text{qFM}\rightarrow\text{qAFM}}^{(2)}$ from the second-order Bertaut expansion	$\propto T_{\text{qFM}\rightarrow\text{qAFM}}^{(2)}$

$$A_{(E_{12}, E_{23})}^R = \frac{i}{\hbar^3} (g_J)^4 \underbrace{\langle E_1 | J_x^{\text{eff}} | E_3 \rangle}_{=M_{13}^{(x)}} \underbrace{\langle E_3 | J_x^{\text{eff}} | E_2 \rangle^*}_{=M_{23}^{(x)*}} \underbrace{\langle E_2 | J_x^{\text{eff}} | E_1 \rangle^*}_{=M_{12}^{(x)*}=0} \underbrace{\langle E_1 | J_x^{\text{eff}} | E_2 \rangle}_{=M_{12}^{(x)}=0}$$

$\implies x\text{-only channel forbidden;}$

$$\begin{aligned} A_{(E_{12}, E_{23})}^{R(\text{odd})} &= \frac{i}{\hbar^3} \mathcal{S}_{12}^{\text{odd}} (g_J h_x)^2 [M_{13}^{(x)}] [M_{23}^{(x)}]^* \frac{e^{-\Gamma_{23}t_2}}{\mathcal{D}_{12}(-\omega_\tau) \mathcal{D}_{23}(\omega_t)}, \\ &= \frac{i}{\hbar^3} \mathcal{S}_{12}^{\text{odd}} (g_J h_x)^2 (2.392)(-0.913) \frac{e^{-\Gamma_{23}t_2}}{\mathcal{D}_{12}^* \mathcal{D}_{23}} \\ &\equiv -\frac{i}{\hbar^3} \mathcal{S}_{12}^{\text{odd}} (g_J h_x)^2 \cdot 2.18 \cdot \frac{e^{-\Gamma_{23}t_2}}{\mathcal{D}_{12}^* \mathcal{D}_{23}}; \end{aligned} \tag{100}$$

$$\begin{aligned} A_{(E_{12}, E_{13})}^R &= \frac{i}{\hbar^3} \mathcal{S}_{12}^{\text{odd}} (g_J h_x)^2 [M_{13}^{(x)}]^2 \frac{e^{-\Gamma_{13}t_2}}{\mathcal{D}_{12}^* \mathcal{D}_{13}} \\ &= \frac{i}{\hbar^3} \mathcal{S}_{12}^{\text{odd}} (g_J h_x)^2 (2.392)^2 \frac{e^{-\Gamma_{13}t_2}}{\mathcal{D}_{12}^* \mathcal{D}_{13}} \\ &\equiv -\frac{i}{\hbar^3} \mathcal{S}_{12}^{\text{odd}} (g_J h_x)^2 \cdot 5.72 \cdot \frac{e^{-\Gamma_{13}t_2}}{\mathcal{D}_{12}^* \mathcal{D}_{13}}; \end{aligned} \tag{101}$$

$$\begin{aligned} A_{(\text{qAFM}, E_{13})}^R &= \frac{i}{\hbar^3} (g_e h_x) [(g_e S) \sin \phi_{\text{qAFM}} \bar{\chi}^{2z}] (g_J h_x)^2 [M_{13}^{(x)}] [M_{12}^{(z)}]^* \\ &\quad \times \langle \text{qAFM} | \delta G_z | \text{vac} \rangle \frac{e^{-\Gamma_{13}t_2}}{\mathcal{D}_{\text{qAFM}}(-\omega_\tau) \mathcal{D}_{13}(\omega_t)}, \end{aligned} \tag{102}$$

with $\sin \phi_{\text{qAFM}} = \mathcal{O}(D_1/J_1)$ the small qAFM δG_z admixture (Sec. XD).

c. Geometry-II peak. VH detection along $\hat{z}$:

$$\begin{aligned} A_{(\text{qAFM}, E_{12})}^R &= \frac{i}{\hbar^3} (g_e h_x)^2 [\sin \phi_{\text{qAFM}} \bar{\chi}^{2z}] (g_J h_z) [M_{12}^{(z)}] [M_{12}^{(z)}]^* \\ &\quad \times \frac{e^{-\Gamma_{12} t_2}}{\mathcal{D}_{\text{qAFM}}(-\omega_\tau) \mathcal{D}_{12}(\omega_t)} \\ &= \frac{i}{\hbar^3} (g_e)^2 g_J h_x^2 h_z \sin \phi_{\text{qAFM}} \bar{\chi}^{2z} (5.264)^2 \frac{e^{-\Gamma_{12} t_2}}{\mathcal{D}_{\text{qAFM}}^* \mathcal{D}_{12}}. \end{aligned} \quad (103)$$

$$\begin{aligned} A_{(E_{12}, \text{qAFM})}^R &= \frac{i}{\hbar^3} (g_J h_z)^2 [M_{12}^{(z)}]^* [M_{12}^{(z)}] \bar{\chi}^{2z} [\sin \phi_{\text{qAFM}}] (g_e h_z) \langle \text{qAFM} | \delta G_z | \text{vac} \rangle \\ &\quad \times \frac{e^{-\Gamma_{\text{qAFM}} t_2}}{\mathcal{D}_{12}(-\omega_\tau) \mathcal{D}_{\text{qAFM}}(\omega_t)}, \end{aligned} \quad (104)$$

the time-reverse of Eq. (103), and

$$\begin{aligned} A_{(\text{qFM}, \text{qAFM})}^R &= \frac{i}{\hbar^3} (g_e h_z)^3 T_{\text{qFM}, \text{qAFM}}^{(2)} (g_e h_x) \langle \text{qFM} | \delta F_z | \text{vac} \rangle \langle \text{qAFM} | \delta G_y | \text{vac} \rangle^2 \\ &\quad \times \frac{e^{-\Gamma_{\text{qAFM}} t_2}}{\mathcal{D}_{\text{qFM}}(-\omega_\tau) \mathcal{D}_{\text{qAFM}}(\omega_t)}, \end{aligned} \quad (105)$$

with $T_{\text{qFM}, \text{qAFM}}^{(2)}$ the effective quadratic mixing rate of Eq. (96), and $\langle m | \delta X | \text{vac} \rangle = \sqrt{S/2} (u_X^m + v_X^m)$ from the Bogoliubov transformation diagonalizing $M_{4 \times 4}^{\Gamma_2}$.

e. Magnitudes and ratios. All six amplitudes share the overall factor $(i/\hbar^3)$ and depend on the same matrix-element catalogue Eq. (98). The cleanest model-independent ratios predicted from the catalogue are

$$\begin{aligned} \frac{A_{(E_{12}, E_{13})}}{A_{(E_{12}, E_{23})}} &= \frac{M_{13}^{(x)}}{M_{23}^{(x)}} = \frac{-2.392 i}{-0.913 i} \approx +2.62, \\ \frac{A_{(\text{qAFM}, E_{12})}^{\text{GeoII}}}{A_{(E_{12}, \text{qAFM})}^{\text{GeoIII}}} &= \frac{h_x^2 h_z \mathcal{D}_{12}^{-1} \mathcal{D}_{\text{qAFM}}^{-1}}{h_z^2 \mathcal{D}_{12}^{-1} \mathcal{D}_{\text{qAFM}}^{-1}} \Big|_{\text{equal } h_x, h_z} = 1, \\ \frac{A_{(\text{qFM}, \text{qAFM})}}{A_{(E_{12}, \text{qAFM})}} &\sim \frac{T^{(2)} (g_e h_z)^2}{(g_J h_z)^2 |M_{12}|^2 \bar{\chi}^{2z}}, \end{aligned} \quad (106)$$

the first of which is parameter-free. The other two reduce the six-amplitude problem to the exchange vertex $\bar{\chi}^{2z}$ and the Fe-sector nonlinear coefficient $T^{(2)}$ once the catalogue Eq. (98) and the linearized magnon sector are fixed.

XIV. PERTURBATIVE EOM DERIVATION OF $\chi^{(3)}$

The Liouville-pathway analysis of Sec. XIII can be re-derived from a complementary, completely classical (Heisenberg) viewpoint by expanding the joint Bertaut+Bloch EOM order by order in the drive amplitude. This provides (i) an independent check of every matrix element in the catalogue Eq. (98), (ii) explicit access to the nonlinear vertices N_2, N_3 that mediate the

cross-sector peaks, and (iii) a clean separation of the Fe-magnon nonlinearities (spin-length constraint, $\text{DM} \times K$ mixing) from the Tm $\text{SU}(3)$ non-Abelian nonlinearities (Gell-Mann commutators).

A. Setup: nonlinear joint EOM

Including all cubic and quartic terms retained in the Hamiltonian, the joint EOM for the 10-component variable $\mathbf{w}$ of Eq. (85) (extended by the two diagonal Tm coordinates $\delta n^3, \delta n^8$ and the four omitted Fe components, $\mathbf{w} \rightarrow \tilde{\mathbf{w}} \in \mathbb{R}^{16}$) has the structure

$$\dot{\tilde{\mathbf{w}}}^a = M^a_b \tilde{\mathbf{w}}^b + N_{2bc}^a \tilde{\mathbf{w}}^b \tilde{\mathbf{w}}^c + N_{3bcd}^a \tilde{\mathbf{w}}^b \tilde{\mathbf{w}}^c \tilde{\mathbf{w}}^d + f^a(t), \quad (107)$$

with:

- M : the linear matrix Eq. (86);
- N_2 : the cubic vertices generated by (a) the $\text{SU}(3)$ commutator $\dot{\rho}_a = -i f_{abc} h_b^{(0)} \rho_c$ entering at the Tm bilinear order, (b) the Bertaut LLG $\mathbf{X} \times \mathbf{H}$ entering at the Fe bilinear order via the spin-length constraint $\mathbf{F}^2 + \mathbf{G}^2 + \mathbf{C}^2 + \mathbf{A}^2 = 4S^2$, and (c) the cross-sector vertex $u_a \lambda^a \mathcal{I}^{\text{Fe}}$ linearized in the Fe coordinate;
- N_3 : the quartic vertices from (a) the $f_{abc} f_{cde} \rho_b \rho_d \rho_e$ Tm triple commutator and (b) the Van Vleck back-action expanded to bilinear in Fe and linear in Tm coherence;

- $f(t)$: the drive force, linear in the field amplitude through the Zeeman and electric-dipole vertices Eqs. (59)–(60).

a. *Explicit form of the nonlinear vertices.* The Tm-only N_2 vertex follows from $\dot{n}^a = f^{abc} h_b^{\text{tot}} n^c$ with $h_b^{\text{tot}} = h_b^{(0)} + h_b^{(1)}(t) + \delta h_b[\mathbf{X}^{\text{Fe}}]$; expanding,

$$\dot{n}^a|_{\mathcal{O}(\tilde{\mathbf{w}}^2)} = N_2^{a,\text{Tm}}{}_{bc} \tilde{\mathbf{w}}^b \tilde{\mathbf{w}}^c = f^{abd} \delta h_b[\tilde{\mathbf{w}}] \delta n_d|_{\mathcal{O}(\tilde{\mathbf{w}}^2)}, \quad (108)$$

which contains the $\tilde{\chi}^{2z} \delta G_z \lambda^2$ vertex contracted by the structure constants f_{2bc} to drive any of the six Tm coherences after projection onto the qAFM admixture. The Fe-only N_2 vertex follows from the bilinear part of the Bertaut precession equation,

$$\dot{\mathbf{X}}|_{\mathcal{O}(\tilde{\mathbf{w}}^2)} = N_2^{\mathbf{X},\text{Fe}}{}_{bc} \tilde{\mathbf{w}}^b \tilde{\mathbf{w}}^c = -\frac{1}{\hbar} \sum_{Y \cdot Z = X} \delta \mathbf{Y} \times \delta \mathbf{H}^Z|_{\mathcal{O}(\tilde{\mathbf{w}}^2)}, \quad (109)$$

$$\begin{aligned} \rho_{12}(t) &\equiv \Lambda_G^2(t) \\ \rho_{13}^{(x)}(t) &\equiv \Lambda_C^5(t), \quad \rho_{13}^{(y)}(t) \equiv \Lambda_A^7(t) \\ \rho_{23}^{(x)}(t) &\equiv \Lambda_C^7(t), \quad \rho_{23}^{(y)}(t) \equiv \Lambda_A^5(t), \\ n_1(t) - n_1^{\text{eq}} &\equiv \Lambda_F^3(t), \quad n_2(t) - n_2^{\text{eq}} \equiv \Lambda_F^8(t), \\ a_{\text{qFM}}(t) &\in \{F_z, G_x\}\text{-block (the } ac\text{-plane } \Gamma_2 \text{ rotation),} \\ a_{\text{qAFM}}(t) &\in \{F_y, G_y\}\text{-block, with } \delta G_z \text{ admixture from DM,} \end{aligned}$$

i.e., the scalar amplitudes $\rho_{\alpha\beta}, n_a, a_m$ should be read as the appropriate Bertaut-projected components labeled by definite $Pbnm$ irreps. Every coupling vertex in Eqs. (116)–(122) is then the contraction of matched σ -vectors guaranteed by the global selection rule Eq. (89): e.g., the $\xi|\rho_{12}|^2$ source on $\dot{a}_{\text{qAFM}}$ closes as $\sigma_G \cdot \sigma_G = \sigma_F$ on the $|\rho_{12}|^2$ side and is projected back onto $G_z \subset \text{qAFM}$, which is what makes the vertex symmetry-allowed. The scalar trees of Sec. XIV C are one-to-one with the six σ -rule rows of Eq. (90), and the peak count is reproduced by symmetry alone.

B. Standard second-order normal-coordinate form

For direct comparison with the magnetic-2DCS literature one may eliminate the conjugate momenta and rewrite the same dynamics in terms of the real coordinates Q_μ of Eq. (66). Extending the linear equation Eq. (67) by the lowest nonlinear terms gives the standard form

$$\ddot{Q}_\mu + 2\Gamma_\mu \dot{Q}_\mu + \Omega_\mu^2 Q_\mu = F_\mu(t) + \sum_{\nu\rho} g_{\mu\nu\rho} Q_\nu Q_\rho + \cdots, \quad (112)$$

and contains, in particular, the qFM–qAFM bilinear coupling $T_{\text{qFM},\text{qAFM}}^{(2)}$ derived in App. E. The cross-sector vertex

$$N_2^{a,\text{Fe} \leftarrow \text{Tm}}{}_{bc} = -\frac{1}{\hbar} \frac{\partial^2 H_{\text{aniso-Tm}}}{\partial \tilde{\mathbf{w}}^b \partial \tilde{\mathbf{w}}^c} \Big|_0 \quad (110)$$

encodes the Van Vleck back-action Eq. (55): setting $b = \delta n^1$ and $c = F_x$ extracts the coefficient $u_1 \alpha_2^{\text{Fe}} F_x$ that appears in the population-induced Fe anisotropy shift.

b. *Bertaut reading of the dynamical variables.* Throughout this section, every Tm dynamical variable is implicitly the *Bertaut combination* Λ_X^a of Eq. (25) selected by the drive (Eqs. (62)) rather than the bare single-site coordinate. Specifically:

$$\begin{aligned} &(\sigma_G\text{-staggered Tm coherence; couples to } h_z, G_z), \\ &(\text{separate } \sigma_C, \sigma_A \text{ projections of the 13 coherence}), \end{aligned} \quad (111)$$

where the nonzero $g_{\mu\nu\rho}$ are fixed by the same symmetry rules as the Liouville pathways. In the present problem the required quadratic vertices are, schematically,

$$\begin{aligned} Q_{\text{qAFM}} &\leftarrow T_{\text{qAFM} \leftrightarrow \text{qFM}}^{(2)} Q_{\text{qFM}} Q_{\text{qAFM}}, \\ Q_{\text{qFM}} &\leftarrow \tilde{T}_{\text{qFM} \leftarrow \text{qAFM}^2}^{(2)} Q_{\text{qAFM}}^2, \\ Q_{12} &\leftarrow g_{12,13,23} Q_{13} Q_{23}, \end{aligned} \quad (113)$$

with $g_{12,13,23} \propto \mathcal{S}_{12}^{\text{odd}}$. Keeping the slow population coordinate $P_{12} \equiv n_1 - n_2$ adds the zero-frequency branch $P_{12} \leftarrow g_{P,12,12} Q_{12}^2$, which is the formal $(E_{12}, 0)$ response discussed below. Equation (112) is the standard normal-coordinate version of the first-order Bloch+LLG system written explicitly in Sec. XIV C.

C. Explicit scalar EOMs in the physical mode basis

It is useful to write Eq. (107) component by component in the diagonal eigenmode basis of M^{Γ^2} . Define the complex mode amplitudes

$$a_{\text{qFM}}(t) = q_{\text{qFM}} + ip_{\text{qFM}}/(\hbar\omega_{\text{qFM}}), \quad (114)$$

$$a_{\text{qAFM}}(t) = q_{\text{qAFM}} + ip_{\text{qAFM}}/(\hbar\omega_{\text{qAFM}}), \quad (115)$$

for the two Fe magnons (so $a \sim e^{-i\omega t}$ at the linear order), together with the six Tm coherences $\rho_{\alpha\beta}(t) \equiv \langle E_\alpha | \hat{\rho} | E_\beta \rangle$ ($\alpha \neq \beta$) and the two independent populations n_1, n_2 (with $n_3 = 1 - n_1 - n_2$). Then Eq. (107) unfolds into the following coupled system, keeping the representative terms needed to expose each peak topol-

ogy in Sec. XIII G. For the electric-dipole channel we use the same shorthand as in Sec. XIII G, namely $D_{12}^{(x)} = \langle E_1 | d_x^{\text{eff}} | E_2 \rangle$, $D_{13}^{(z)} = \langle E_1 | d_z^{\text{eff}} | E_3 \rangle$, and $D_{23}^{(z)} = \langle E_2 | d_z^{\text{eff}} | E_3 \rangle$:

(i) *Fe quasi-FM magnon*. Drive-only at first order; at second order the Tm population n_1 acts as a static restoring-field shift through the Van Vleck channel:

$$\dot{a}_{\text{qFM}} = -(i\omega_{\text{qFM}} + \Gamma_{\text{qFM}}) a_{\text{qFM}} + \underbrace{\frac{g_e \mu_B}{\hbar} h_z(t)}_{\text{Zeeman drive}} + \underbrace{T_{\text{qFM,qAFM}}^{(2)} a_{\text{qAFM}}^2}_{\text{spin-length}} + \underbrace{\zeta_1 n_1 a_{\text{qFM}}}_{\text{Van Vleck shift}}. \quad (116)$$

(ii) *Fe quasi-AFM magnon*. The DM-bilinear $T^{(2)}$ vertex feeds qFM into qAFM and vice versa; the Tm population

n_1 both shifts and rotates the qAFM mode (because Van Vleck modifies K_2^{Fe} , which is what gives qAFM its DM-protected gap):

$$\dot{a}_{\text{qAFM}} = -(i\omega_{\text{qAFM}} + \Gamma_{\text{qAFM}}) a_{\text{qAFM}} + \sin \phi_{\text{qAFM}} \frac{g_e \mu_B}{\hbar} h_x(t) + T_{\text{qAFM,qFM}}^{(2)} a_{\text{qFM}} a_{\text{qAFM}} + \zeta_2 n_1 a_{\text{qAFM}} + \xi |\rho_{12}|^2. \quad (117)$$

The very last term, bilinear in the Tm coherence, is the source of the (E_{12}, qAFM) rephasing peak: a coherent grating in ρ_{12} at frequency $\omega_2 - \omega_1$ launches a qAFM magnon at the same frequency through the cross vertex $\xi = \partial^2 H_{\text{aniso-Tm}} / \partial |\rho_{12}|^2 \partial G_z|_0$.

(iii) *Tm $|1\rangle \leftrightarrow |2\rangle$ coherence*. Driven by h_z via $M_{12}^{(z)}$; modulated nonlinearly by Fe magnons through δG_z and by populations through SU(3) symmetric vertex:

$$\dot{\rho}_{12} = -(i\omega_{12} + \Gamma_{12}) \rho_{12} + \frac{i}{\hbar} M_{12}^{(z)} g_J h_z(t) (n_1 - n_2) + \frac{i}{\hbar} D_{12}^{(x)} e_x(t) (n_1 - n_2) + \frac{i}{\hbar} \bar{\chi}^{2z} \delta G_z(t) (n_1 - n_2) + \frac{i}{\hbar} [M_{13}^{(x)} g_J h_x(t) + D_{13}^{(z)} e_z(t)] (n_1 - n_3) + \frac{i}{\hbar} [M_{23}^{(x)} g_J h_x(t) + D_{23}^{(z)} e_z(t)] (n_2 - n_3). \quad (118)$$

The h_x/e_z odd-sector rotation terms are the Liouville R -pathway junctions that rotate $\rho_{13} \rightarrow \rho_{12}$ during t_2 and

produce the (E_{12}, E_{23}) and (E_{12}, E_{13}) peaks.

(iv) *Tm $|1\rangle \leftrightarrow |3\rangle$ and $|2\rangle \leftrightarrow |3\rangle$ coherences*.

$$\dot{\rho}_{13} = -(i\omega_{13} + \Gamma_{13}) \rho_{13} + \frac{i}{\hbar} [M_{13}^{(x)} g_J h_x + M_{13}^{(y)} g_J h_y + D_{13}^{(z)} e_z] (n_1 - n_3) + \frac{i}{\hbar} [M_{12}^{(z)} g_J h_z + D_{12}^{(x)} e_x] \rho_{23} - \frac{i}{\hbar} [M_{23}^{(x)*} g_J h_x + D_{23}^{(z)*} e_z] \rho_{13} + \frac{i}{\hbar} [M_{23}^{(y)*} g_J h_y + D_{23}^{(z)*} e_z] \rho_{13}. \quad (119)$$

$$\dot{\rho}_{23} = -(i\omega_{23} + \Gamma_{23}) \rho_{23} + \frac{i}{\hbar} [M_{23}^{(x)} g_J h_x + M_{23}^{(y)} g_J h_y + D_{23}^{(z)} e_z] (n_2 - n_3) + \frac{i}{\hbar} [M_{12}^{(z)*} g_J h_z + D_{12}^{(x)*} e_x] \rho_{13} - \frac{i}{\hbar} [M_{13}^{(x)*} g_J h_x + D_{13}^{(z)*} e_z] \rho_{23} + \frac{i}{\hbar} [M_{13}^{(y)*} g_J h_y + D_{13}^{(z)*} e_z] \rho_{23}. \quad (120)$$

(v) *Tm populations*. Driven only at second order, forming

the diagonal SU(3) gratings that mediate every cross-sector peak:

$$\dot{n}_1 = -\Gamma_1^{\text{pop}}(n_1 - n_1^{\text{eq}}) + \frac{2}{\hbar} \text{Im}[M_{12}^{(z)*} g_J h_z(t) \rho_{12}] + \frac{2}{\hbar} \text{Im}[D_{12}^{(x)*} e_x(t) \rho_{12}] + \frac{2}{\hbar} \text{Im}[(M_{13}^{(x)*} g_J h_x(t) + D_{13}^{(z)*} e_z(t)) \rho_{13}], \quad (121)$$

$$\dot{n}_2 = -\Gamma_2^{\text{pop}}(n_2 - n_2^{\text{eq}}) - \frac{2}{\hbar} \text{Im}[M_{12}^{(z)*} g_J h_z(t) \rho_{12}] + \frac{2}{\hbar} \text{Im}[D_{12}^{(x)*} e_x(t) \rho_{12}] + \frac{2}{\hbar} \text{Im}[(M_{23}^{(x)*} g_J h_x(t) + D_{23}^{(z)*} e_z(t)) \rho_{23}]. \quad (122)$$

The h_z/e_x source on $\dot{n}_1$ and $\dot{n}_2$ is precisely the second-order vertex that creates the δn^1 grating used in the Van Vleck cascade $\zeta_1 n_1 a_{\text{qAFM}}$ of Eq. (116) and $\xi |\rho_{12}|^2$ of Eq. (117). The same source is the formal origin of the population branch $(E_{12}, 0)$ discussed in Sec. XIII.

Reading off the six peaks. Iterating Eqs. (116)–(122) to third order in the drive generates exactly the six nonlinear processes (one per peak in Sec. XIII G). Each peak is identified by its tree topology in the scalar EOM:

Peak	Tree in Eqs. (116)–(122)
(E_{12}, E_{23})	$\{h_x J_x^{\text{eff}}, e_z d_z^{\text{eff}}\}^{\otimes 2} \rightarrow \rho_{12}^{\text{odd}}$; then odd-sector readout at ω_{23}
(E_{12}, E_{13})	same ρ_{12}^{odd} source, followed by odd-sector readout at ω_{13}
(E_{12}, qAFM)	$(h_z, e_x) \rightarrow \rho_{12}$; then $\bar{\chi}^{2z} \rho_{12} \rightarrow \delta G_z \rightarrow a_{\text{qAFM}}$ through the qAFM admixture
(qAFM, E_{12})	$h_x \rightarrow a_{\text{qAFM}} \rightarrow \delta G_z$; then $\bar{\chi}^{2z} \delta G_z (n_1 - n_2)$ in (118)
(qAFM, E_{13})	$h_x \rightarrow a_{\text{qAFM}} \rightarrow \delta G_z$; then $\bar{\chi}^{2z} \delta G_z \rightarrow \rho_{12}$, followed by odd-sector rotation into ρ_{13}
$(\text{qFM}, \text{qAFM})$	$h_z \rightarrow a_{\text{qFM}}$, followed by the effective Fe-only quadratic mixing vertex $T^{(2)}$ before qAFM readout

In this language the $\dot{a}_{\text{qAFM}}$ EOM contains three distinct nonlinear sources — the effective Fe-only vertex $T^{(2)}$, $\zeta_2 n_1 a_{\text{qAFM}}$, and $\xi |\rho_{12}|^2$ — each one a different 2DCS peak; and the $\dot{\rho}_{12}$ EOM contains four ($M^{(z)} h_z (n_1 - n_2)$, $\bar{\chi}^{2z} \delta G_z (n_1 - n_2)$, and the two SU(3) raising terms $h_x \rho_{32}$, $h_x \rho_{13}$). The total count of independent trilinear terms across Eqs. (116)–(122) is six — matching the six peaks of Table III.

In Fourier space with the resolvent (Green's function) $G(\omega) \equiv (i\omega \mathbf{1} - M)^{-1}$,

$$\begin{aligned} \tilde{\mathbf{w}}^{(1)}(\omega_1) &\equiv G(\omega_1) f^{(1)}(\omega_1), \\ \tilde{\mathbf{w}}^{(2)}(\omega_1 + \omega_2) &\equiv G(\omega_1 + \omega_2) N_2[\tilde{\mathbf{w}}^{(1)}(\omega_1), \tilde{\mathbf{w}}^{(1)}(\omega_2)], \\ \tilde{\mathbf{w}}^{(3)}(\Omega) &\equiv G(\Omega) \left\{ 2N_2[\tilde{\mathbf{w}}^{(1)}, \tilde{\mathbf{w}}^{(2)}] \right. \\ &\quad \left. + N_3[\tilde{\mathbf{w}}^{(1)}, \tilde{\mathbf{w}}^{(1)}, \tilde{\mathbf{w}}^{(1)}] \right\}, \end{aligned} \quad (126)$$

with $\Omega = \omega_1 + \omega_2 + \omega_3$. The detected polarization $P_{\text{det}}^{(3)}(t) = \langle \hat{O}_{\text{det}}, \tilde{\mathbf{w}}^{(3)}(t) \rangle$ gives the third-order signal, and $\chi^{(3)}(\omega_t, \omega_2, \omega_\tau)$ is read off by Fourier transform with $\omega_1 = \pm\omega_\tau$, ω_2 slow, $\omega_3 = \omega_t - \omega_1 - \omega_2$.

D. Order-by-order expansion

Writing $f(t) = \lambda f^{(1)}(t)$ for a formal small parameter λ counting powers of the drive, expand $\tilde{\mathbf{w}} = \sum_{n \geq 1} \lambda^n \tilde{\mathbf{w}}^{(n)}$. Substituting in Eq. (107) and collecting powers:

$$\dot{\tilde{\mathbf{w}}}^{(1)} = M \tilde{\mathbf{w}}^{(1)} + f^{(1)}(t), \quad (123)$$

$$\dot{\tilde{\mathbf{w}}}^{(2)} = M \tilde{\mathbf{w}}^{(2)} + N_2[\tilde{\mathbf{w}}^{(1)}, \tilde{\mathbf{w}}^{(1)}], \quad (124)$$

$$\dot{\tilde{\mathbf{w}}}^{(3)} = M \tilde{\mathbf{w}}^{(3)} + 2N_2[\tilde{\mathbf{w}}^{(1)}, \tilde{\mathbf{w}}^{(2)}] + N_3[\tilde{\mathbf{w}}^{(1)}, \tilde{\mathbf{w}}^{(1)}, \tilde{\mathbf{w}}^{(1)}]. \quad (125)$$

E. Numerical 2DCS simulation protocol

For comparison with experiment one solves the full nonlinear equations Eq. (107) (or the reduced normal-coordinate form Eq. (112)) three times with identical pulse envelopes and damping parameters: once with both pulses present, once with pulse A only, and once with pulse B only. The simulated nonlinear field is then constructed by the same subtraction used experimentally, Eq. (65), and Fourier transformed as

$$S_{\text{sim}}(\omega_t, t_2, \omega_\tau) = \int_0^\infty dt_3 e^{i\omega_t t_3} \int_0^\infty dt_1 e^{\pm i\omega_\tau t_1} E_{\text{sig}}^{\text{NL}}(t_3, t_2, t_1). \quad (127)$$

In practice Eq. (63) is the microscopic starting point for Monte Carlo or bond-resolved parameter sampling, while Eqs. (64), (67), and (112) provide the reduced driven dynamics whose peak positions, amplitudes, and geometry selectivity are compared against Table III.

F. Structure of $G(\omega)$

The resolvent factorizes into uncoupled blocks at lowest order in D/J :

$$G(\omega) = G^{\text{Fe}}(\omega) \oplus G^{\text{Tm}}(\omega) + \mathcal{O}(D_1/J_1), \quad (128)$$

with the Tm block diagonal in the three coherence pairs,

$$G_{\alpha\beta \rightarrow \alpha\beta}^{\text{Tm}}(\omega) = \frac{1}{i(\omega - \omega_{\alpha\beta}) + \Gamma_{\alpha\beta}} \equiv \mathcal{D}_{\alpha\beta}^{-1}(\omega), \quad (129)$$

which are exactly the coherence denominators of Eq. (97). The Fe block, after Bogoliubov diagonalization of $M_{4 \times 4}^{\Gamma_2}$, becomes diagonal in the {qFM, qAFM} eigenmodes with denominators $\mathcal{D}_{\text{qFM, qAFM}}(\omega)$. The off-block correction $\mathcal{O}(D_1/J_1)$ produces the $\sin \phi_{\text{qAFM}} \sim D_1/J_1$ admixture used throughout Sec. XIII.

$$P_{\lambda^2}^{(3)}(\Omega) = [M_{12}^{(z)}] G_{12}^{\text{Tm}}(\Omega) \bar{\chi}^{2z} \sin \phi_{\text{qAFM}} G_{\text{qAFM}}^{\text{Fe}}(\omega_1) (g_e h_x) (g_J h_z) \cdot (g_e h_x). \quad (131)$$

Up to the universal $(i/\hbar^3)$ this is identical to Eq. (103).

c. Fe-only nonlinearity — pathway (qFM, qAFM). Pure N_2^{Fe} in two cascades: $\tilde{\mathbf{w}}_{\text{qFM}}^{(1)} \rightarrow \tilde{\mathbf{w}}_{\text{qAFM}}^{(2)}$ via the

$$\tilde{\mathbf{w}}_{\text{qAFM}}^{(2)}(\omega_1 + \omega_2) = G_{\text{qAFM}}^{\text{Fe}}(\omega_1 + \omega_2) T_{\text{qFM, qAFM}}^{(2)} \tilde{\mathbf{w}}_{\text{qFM}}^{(1)}(\omega_1) \tilde{\mathbf{w}}_{\text{qAFM}}^{(1)}(\omega_2), \quad (132)$$

followed by one more drive interaction at order λ^3 to give the detected polarization. This is the EOM realization of Eq. (105) and exposes the spin-length-constraint origin of the $T^{(2)}$ vertex (the $|\mathbf{S}| = S$ sphere curvature).

$$G_{\alpha\beta}^{\text{Tm}}(\omega) \longleftrightarrow \mathcal{G}_{|E_\alpha\rangle\langle E_\beta|}(\omega), \quad N_2, N_3 \longleftrightarrow \text{double-sided Feynman vertices}, \quad (133)$$

together with the identification of f_{abc}, d_{abc} structure constants with the antisymmetric/symmetric parts of the $\hat{V}_i$ vertices in Eq. (88). Every A_{peak} in Sec. XIII G arises from one tree diagram in Eqs. (123)–(125); the tree topology is fixed by the Liouville pathway. This consistency

G. Pathway-by-pathway recovery of $\chi^{(3)}$

We now show that each Liouville pathway of Sec. XIII arises from a unique tree diagram in Eqs. (123)–(125), and that the matrix-element products are the same as in Sec. XIII G.

a. Tm-only $\chi^{(3)}$ — pathway (E_{12}, E_{23}). Pure odd-sector SU(3) cascade. Two first-order Geometry-I odd-sector drives on $\lambda^{4,5,6,7}$ generate an effective source $\mathcal{S}_{12}^{\text{odd}}$ on the $\lambda^{1,2}$ sector; a third odd-sector vertex then reads out the 23 coherence. Schematically,

$$\tilde{\mathbf{w}}_{\lambda^7}^{(3)}(\Omega) \neq G_{23}^{\text{Tm}}(\Omega) \mathcal{S}_{12}^{\text{odd}}(\omega_1, \omega_2) \tilde{\mathbf{w}}_{\lambda^7}^{(1)}(\omega_3), \quad (130)$$

where $\mathcal{S}_{12}^{\text{odd}}$ collects the allowed odd-sector bilinears (4, 6), (5, 7), (4, 7), and (5, 6) that close onto $\lambda^{1,2}$. This yields the denominator structure and matrix-element bookkeeping summarized in Eq. (100).

b. Fe $\rightarrow$ Tm exchange — pathway (qAFM, E_{12}). Two-step diagram: (i) $\tilde{\mathbf{w}}_{\delta G_z}^{(1)} = G_{\text{qAFM}}^{\text{Fe}}(\omega_1) \sin \phi_{\text{qAFM}} (g_e h_x)$ (magnon excited then projected onto δG_z); (ii) $\tilde{\mathbf{w}}_{\lambda^2}^{(2)} = G_{12}^{\text{Tm}}(\omega_1 + \omega_2) \bar{\chi}^{2z} \tilde{\mathbf{w}}_{\delta G_z}^{(1)}(\omega_1) (g_J h_z)$; (iii) detection at $\omega_t = \omega_{12}$ gives

$T_{\text{qFM, qAFM}}^{(2)}$ vertex. Explicitly,

H. Summary: EOM-Liouville dictionary

The dictionary between the two derivations is the identification

between the operator-algebra (Liouville) and classical-EOM viewpoints is the manuscript's strongest internal cross-check that the six peaks have been assigned to physically allowed nonlinearities of H_{total} .

XV. CONCLUSION

We have constructed a symmetry-allowed minimal Hamiltonian for TmFeO₃ in the Γ_2 phase, addressing the six peaks of the corrected 2DCS target list through explicit Liouville-pathway derivations. The essential physics ingredients are:

1. the four-sublattice Fe LLG sector with proper DM-gapped qAFM frequency, written via an effective reduced 4×4 linearized magnon matrix M^{Γ_2} (Sec. XB);
2. the three Tm singlets in the T -symmetric gauge with explicit $m_J \rightarrow -m_J$ parity assignments (Sec. III);
3. the surviving uniform Fe–Tm exchange vertex $\bar{\chi}^{2z} \delta G_z \lambda^2$ after inversion-pair cancellation of all mirror-odd channels at $\mathbf{q} = 0$ (Sec. V);
4. the Van Vleck back-action through the A_1^+ Tm density coordinates $\lambda^{1,3,8}$, which controls the population-side nonlinear feedback entering the Fe sector (Sec. VD);
5. a CPTP-compliant Lindblad dissipator (Sec. IX);
6. a coherent emission expression with the SI impedance prefactor on the magnetic-dipole channel (Sec. XI);
7. a Liouville-pathway derivation that gives every peak in Tab. III a definite amplitude formula in terms of microscopic matrix elements and reduced-model nonlinear vertices (Sec. XIII).

The model is restricted to the low-temperature regime $T \lesssim 20$ K where the linearization around $|E_1\rangle$ is justified. Extension to the spin-reorientation window $T_{\text{SR1}} < T < T_{\text{SR2}}$ requires re-linearization around a thermally mixed reference state and is left for future work, as is the explicit treatment of magnon-polariton dispersion in the biaxial host.

Appendix A: Conventions glossary

A single-page reference for every prefactor used in the manuscript.

- Gell-Mann normalization: $\text{Tr}(\lambda^a \lambda^b) = 2\delta^{ab}$, $[\lambda^a, \lambda^b] = 2if^{abc}\lambda^c$, $\{\lambda^a, \lambda^b\} = \frac{4}{3}\delta^{ab} + 2d^{abc}\lambda^c$.

- Hamiltonian on Tm qutrit: $H = h_0 \mathbb{1} + \frac{1}{2} \sum_a h^a \lambda^a$, hence $h^a = \text{Tr}(H \lambda^a)$ and the SU(3) Bloch EOM is $\dot{n}^a = \sum_{bc} f^{abc} h^b n^c$.
- Bertaut definition: $\mathbf{X} = \frac{1}{4} \sum_\mu \sigma_\mu^X \mathbf{S}_\mu$, inverse $\mathbf{S}_\mu = \sum_X \sigma_\mu^X \mathbf{X}$, orthogonality $\sum_\mu \sigma_\mu^X \sigma_\mu^Y = 4\delta^{XY}$.
- Per-site effective field on Fe: $\mathbf{H}_{\mathbf{n},\mu}^{\text{eff}} = -\partial H_{\text{Fe}} / \partial \mathbf{S}_{\mathbf{n},\mu}$. Bertaut-projected field: $\mathbf{H}_{\mathbf{n},\mu}^{X,\text{site}} = \frac{1}{4} \sum_\mu \sigma_\mu^X \mathbf{H}_\mu^{\text{eff}}$.
- Anisotropy: $H_{\text{aniso}} = -\sum_{\mathbf{n},\mu} \sum_a K_a (S_{\mathbf{n},\mu}^a)^2$, giving $\mathbf{H}_{\mathbf{n},\mu}^{\text{eff},\text{aniso}} = 2\mathbf{K} \mathbf{S}_\mu$ and $\mathbf{H}_{\text{aniso}}^{X,\text{site}} = 2\mathbf{K} \mathbf{X}$ (no extra factor of 4).
- Exchange: $H_{\text{Heis}} = \sum_{\langle ij \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$, giving the per-site $\mathbf{H}_{\mathbf{n},\mu}^{\text{eff},\text{Heis}} = -2 \sum_j J_{\mu j} \mathbf{S}_j$ and the Bertaut $\mathbf{H}_{\text{Heis}}^{X,\text{site}} = -2J^X \mathbf{X}$ with J^X from Eq. (45).
- Larmor with energy-valued effective field: $\dot{\mathbf{S}} = -\hbar^{-1} \mathbf{S} \times \mathbf{H}^{\text{eff}}$. For a Zeeman term $-g_e \mu_B \mathbf{h} \cdot \mathbf{S}$ this becomes $\dot{\mathbf{S}} = -(g_e \mu_B / \hbar) \mathbf{S} \times \mathbf{h} \equiv -\gamma_e \mathbf{S} \times \mathbf{h}$.
- Magnetic moment of Tm³⁺: $\boldsymbol{\mu} = -g_J \mu_B \mathbf{J}$ with $g_J = 7/6$.
- Stevens parameters in Tab. I: Hutchings convention with Stevens factors absorbed.

Appendix B: DM Bertaut prefactor derivation

With the bond enumeration of App. D (4 in-plane NN bonds per cell at $z = \frac{1}{2}$ with $\mathbf{D} = (0, +D_1, +D_2)$ and 4 at $z = 0$ with $\mathbf{D} = (0, -D_1, +D_2)$), direct symbolic projection of $\sum_{\langle ij \rangle} \mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j)$ onto the Bertaut modes (Sec. IV C) yields

a. Explicit antisymmetrization. For a single bond $\mu \rightarrow \nu$ in the Bertaut basis, $(\mathbf{S}_\mu \times \mathbf{S}_\nu)_a = \sum_{X,Y} \sigma_\mu^X \sigma_\nu^Y (\mathbf{X} \times \mathbf{Y})_a = \frac{1}{2} \sum_{X \neq Y} (\sigma_\mu^X \sigma_\nu^Y - \sigma_\mu^Y \sigma_\nu^X) (\mathbf{X} \times \mathbf{Y})_a$. With the four Fe₁ $\rightarrow$ Fe₂ bonds in $z = \frac{1}{2}$ the nonzero antisymmetric Bertaut differences are $(\sigma_1^F \sigma_2^G - \sigma_1^G \sigma_2^F) = -2$, $(\sigma_1^F \sigma_2^A - \sigma_1^A \sigma_2^F) = -2$, $(\sigma_1^G \sigma_2^C - \sigma_1^C \sigma_2^G) = +2$, $(\sigma_1^C \sigma_2^A - \sigma_1^A \sigma_2^C) = -2$; for Fe₃ $\rightarrow$ Fe₄ in $z = 0$ the corresponding differences are $+2, -2, +2, +2$. Summing $4(\mathbf{S}_1 \times \mathbf{S}_2) + 4(\mathbf{S}_3 \times \mathbf{S}_4)$ with $\mathbf{D} = (0, +D_1, +D_2)$ and $\mathbf{D}' = (0, -D_1, +D_2)$ respectively gives

$$H_{\text{DM}}/N_c = -8D_1 [(\mathbf{F} \times \mathbf{G})_y + (\mathbf{C} \times \mathbf{A})_y] + 8D_2 [(\mathbf{G} \times \mathbf{C})_z - (\mathbf{F} \times \mathbf{A})_z]. \quad (\text{B1})$$

Key structural consequences:

- The D_1 invariant $(\mathbf{F} \times \mathbf{G})_y = F_z G_x - F_x G_z$ reduces to $-F_x G_z$ on the Γ_2 background, recovering the canonical Treves–Bertaut weak-FM Lifshitz en-

ergy $+8N_c D_1 F_x G_z$ and the standard canting $F_x/G_z \simeq D_1/H_E^{\text{exch}}$ [Eq. (49)]. The partner $(\mathbf{C} \times \mathbf{A})_y$ is its Γ_3 -channel image.

- The D_2 component generates $\mathbf{G} \times \mathbf{C}$ and $\mathbf{F} \times \mathbf{A}$ (the

latter with a relative minus). $(\mathbf{G} \times \mathbf{C})_z = G_x C_y - G_y C_x$ reduces to zero on the Γ_2 background and is the small $\mathcal{O}(D_2 C_y / H_E)$ correction; it does not affect the leading weak-FM picture.

- The overall prefactor and sign are those of the directed-bond convention used in Sec. IV and the implementation in `src/core/unitcell_builders.cpp`; reversing every bond direction flips the overall sign but not the invariant content.

Appendix C: Inversion-pair cancellation of mirror-odd Fe–Tm channels

Each Fe–Tm bond shell contains 8 bonds organized into 4 inversion-pair orbits (App. D). Inversion at the bond midpoint sends the Tm site j to its inversion partner $\sigma_I^{\text{Tm}}(j) = (14)(23)(j)$ and the Fe site to itself. Under this inversion, the spin $\mathbf{S}_i$ at Fe_i is unchanged (axial), but the Tm bilinear operator λ^a transforms according to its $A_{1,2}^\pm$ class (Eq. (22)):

$$\eta_a = \begin{cases} +1, & a \in \{1, 2, 3, 8\} \text{ (mirror-even)}, \\ -1, & a \in \{4, 5, 6, 7\} \text{ (mirror-odd)}, \end{cases} \quad (\text{C1})$$

so that $I : \lambda^a \rightarrow \eta_a \lambda^a$. For a pair of inversion-related bonds, the on-site $\mathbf{q} = 0$ projection $\sum_{\text{pair}} \chi^{ab} \lambda^a S^b$ becomes $(\eta_a + 1) \chi^{ab} \lambda^a S^b$ at $\mathbf{q} = 0$, which vanishes when $\eta_a = -1$. Hence:

- all $\lambda^{4,5,6,7}$ on-site couplings to Fe spins cancel at $\mathbf{q} = 0$;
- only $\lambda^{1,2,3,8}$ on-site couplings survive.

This is the symmetry origin of the dominant $\bar{\chi}^{2z}$ vertex.

Appendix D: Bond enumeration

1. Fe–Fe NN, NNN, and out-of-plane bonds

a. In-plane NN (J_1). Each Fe sublattice has 4 in-plane NN neighbors, distributed between two sublattices

via the S_1 screw. The in-plane NN bonds form two sets: those in the $z = \frac{1}{2}$ plane connecting $\text{Fe}_1 \leftrightarrow \text{Fe}_2$, and those in $z = 0$ connecting $\text{Fe}_3 \leftrightarrow \text{Fe}_4$. Coordination $z_1 = 4$.

b. Out-of-plane NN (J_{1c}). Each Fe sublattice has 2 out-of-plane NN neighbors connecting $z = 0$ and $z = \frac{1}{2}$ planes ($\text{Fe}_1 \leftrightarrow \text{Fe}_4$, $\text{Fe}_2 \leftrightarrow \text{Fe}_3$). Coordination $z_{1c} = 2$.

c. NNN (J_2). Within each sublattice, 6 NNN neighbors $(\pm\hat{a}, \pm\hat{b}, \pm\hat{c})$. $z_2 = 6$.

2. Fe–Tm bond shells

Brute-force enumeration of all Fe–Tm pairs within a 4 Å cutoff gives four distance shells:

$$\begin{aligned} d_1 &= 3.054 \text{ Å}, & d_2 &= 3.179 \text{ Å}, \\ d_3 &= 3.357 \text{ Å}, & d_4 &= 3.711 \text{ Å}. \end{aligned} \quad (\text{D1})$$

Each shell has 8 bonds per cell organized in 4 inversion pairs. The distance spread of 21% means the Fe–Tm exchange parameters scale significantly between shells; we parametrize the radial dependence as $\chi(d) \propto d^{-n}$ with $n \in \{6, 8, 10\}$ in the simulation.

Appendix E: Second-order Bertaut expansion

The bilinear couplings entering Eq. (96) are derived from the spin-length constraint and the DM-induced cross-terms in the LLG expansion. At second order in the linearization $\mathbf{X} = \bar{\mathbf{X}} + \delta\mathbf{X}$, the cross-product structure $\delta\mathbf{X}^{(2)} = -\hbar^{-1} \sum_{Y \cdot Z = X} \delta\mathbf{Y} \times \delta\mathbf{H}^Z$ generates the qFM–qAFM coupling. A representative contribution is

$$\delta\dot{F}_z^{(2)} \supset -(D_1/\hbar) \delta F_y \delta G_y, \quad (\text{E1})$$

which projects onto the effective reduced-mode coefficient $T_{\text{qFM,qAFM}}^{(2)}$ of Eq. (96). This appendix is therefore a scaling diagnostic for the Fe-only quadratic vertex rather than a closed-form formula for the reduced-mode coefficient itself.

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Appendix A: Executable verification status

The current repository does *not* contain a single monolithic companion script named `verify_tmfeo3.py`. Instead, the executable checks used in this audit are dis-

tributed across several utilities, and their coverage is only partial. The present status is:

1. `util/verify_tmfeo3_code.py` checks the implemented crystal structure, local-frame construction, Fe-Fe exchange and DM matrices, Fe-Tm bond tables, Tm sublattice frames, bare h_3/h_8 conventions, and the on-site/inter-site trilinear sign structure against the notes.
2. `util/verify_tmfeo3_bonds.py` checks the Fe-Fe, Fe-Tm, and Tm-Tm bond enumerations, the inversion-pair organization of the Fe-Tm shells, and the DM sign convention between the $z = \frac{1}{2}$ and $z = 0$ planes.
3. `util/verify_tmfeo3_eom.py` verifies the Gell-Mann algebra, Bertaut Klein-four identities, and a symbolic EOM expansion, but it is written in the alternate convention $\sigma^G = (+, -, +, -)$, $\sigma^A = (+, -, -, +)$. Its output therefore cannot be compared line-by-line to the corrected manuscript without a convention translation.
4. The Sec. III qutrit spectrum, the corrected Stevens table, and the projected J_a matrices were verified separately during this audit by direct diagonalization, but that check is not yet packaged as a repository utility.
5. The corrected DM reduction of App. B and the projected $\bar{\chi}^{2z} \sin \phi_{\text{qAFM}}$ coupling of Eq. (87) are therefore supported by a combination of code audit, bond-enumeration checks, and local symbolic reduction rather than by one all-encompassing script.
6. The CPTP Lindblad form of Sec. IX is now specified analytically in a complete projector-based form, but the repository does not yet contain a dedicated executable Choi-positivity check for that exact manuscript parameterization.

The corrected manuscript should therefore be read as supported by a set of partial executable checks plus local symbolic derivations, not by a single monolithic verification script.